PI System Client

SUTTHIPHAT SAENCHAIYA

Date 27 May 2019

Day 1

- PI System Basics
- How Data Flows in the PI System
- Basics PI AF
- PI Vision

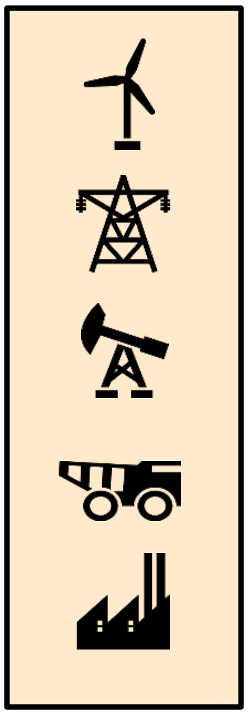
Day 2

- PI ProcessBook
- PI DataLink

PI System Basics

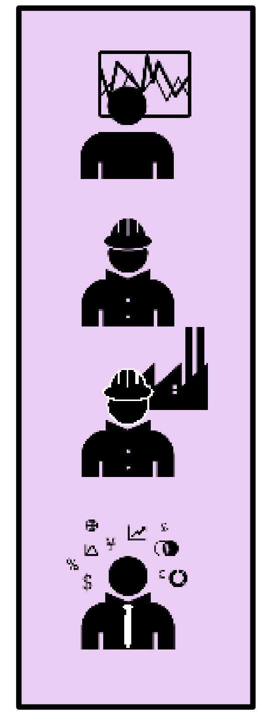
What is the PI System?

Data sources

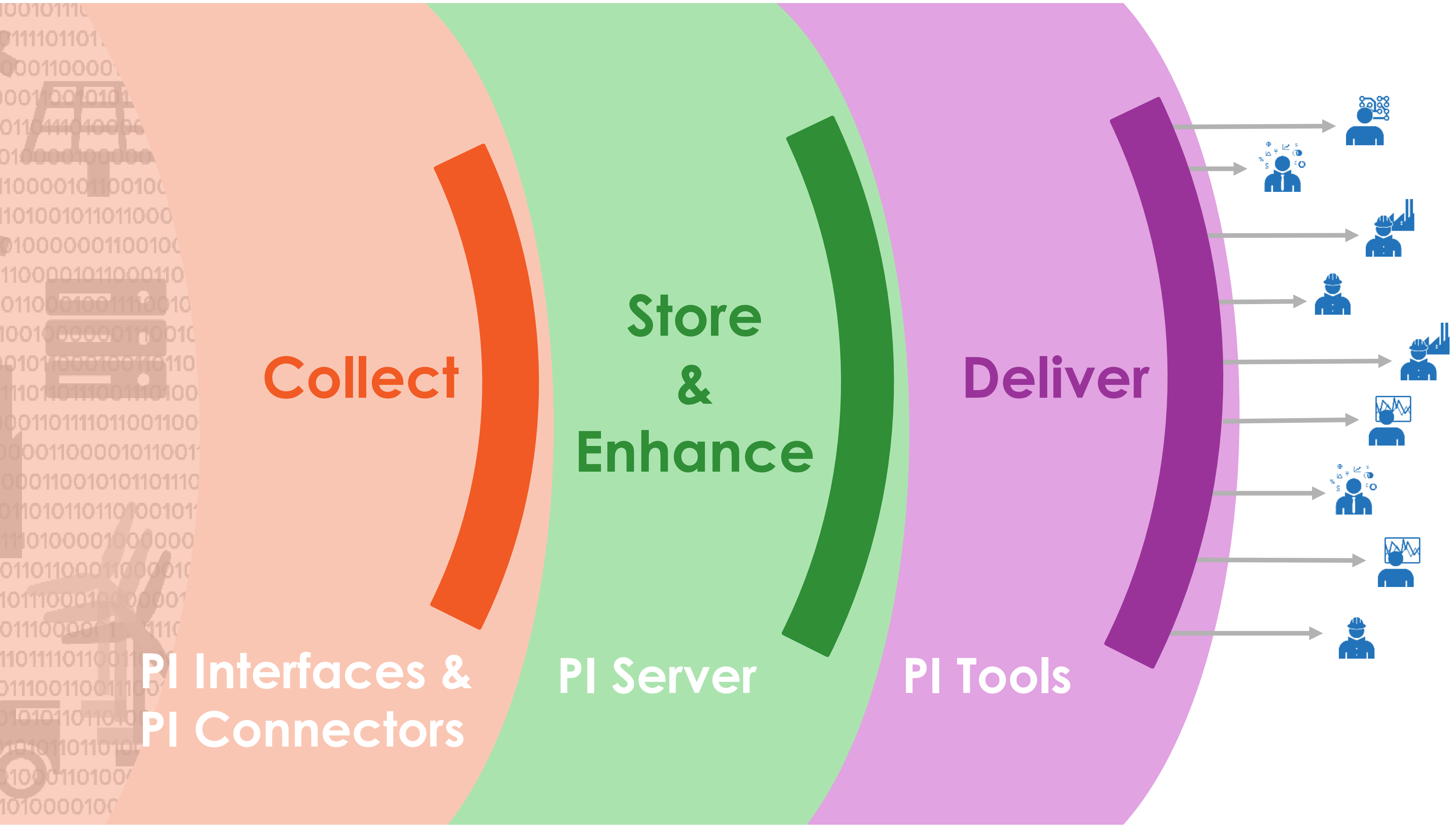


The **PI System** is all the software in between

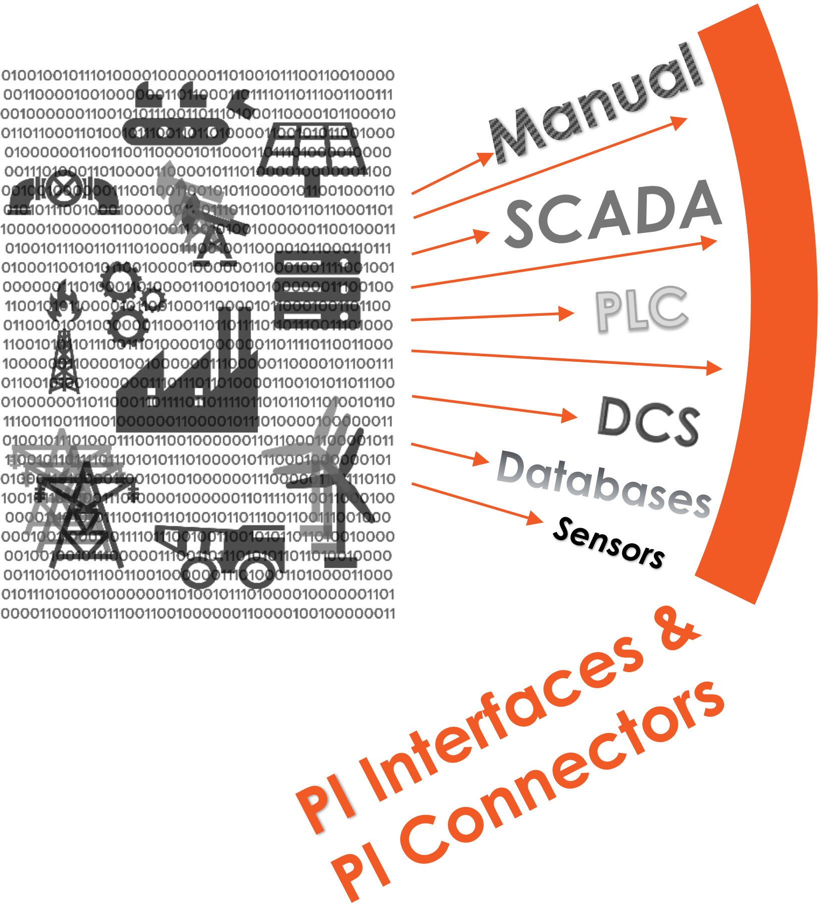
Data Consumers



The PI System has 3 Layers



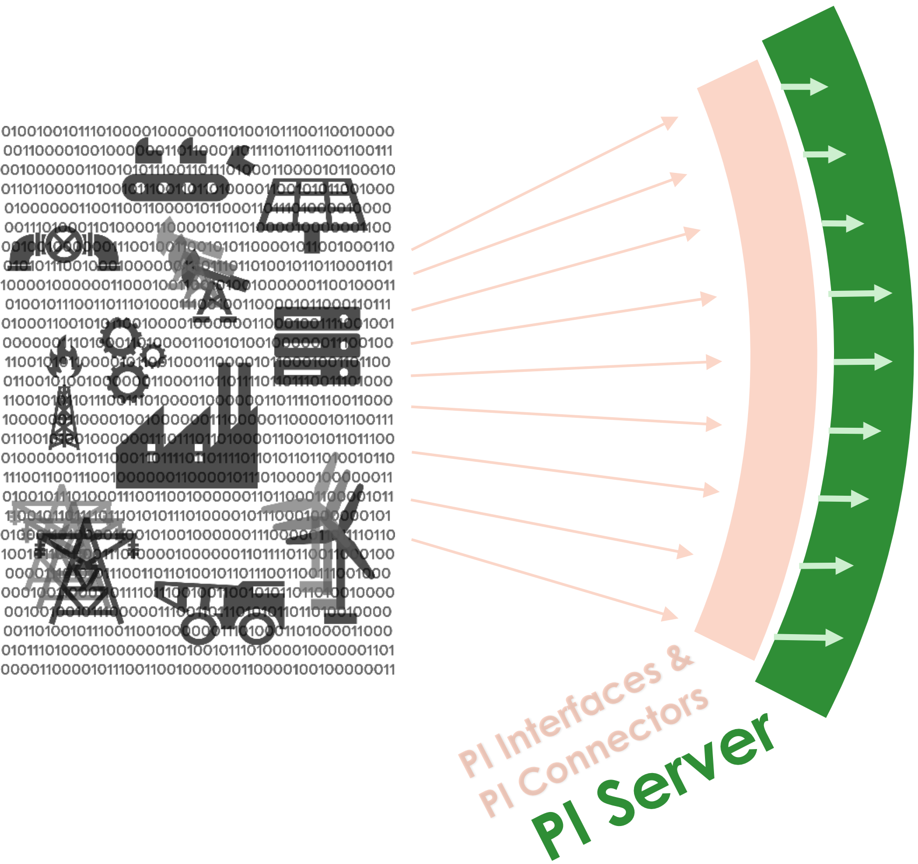
1. Collect



Collects time-series data from any source…

We have over 300 data collection applications

2. Store and Enhance

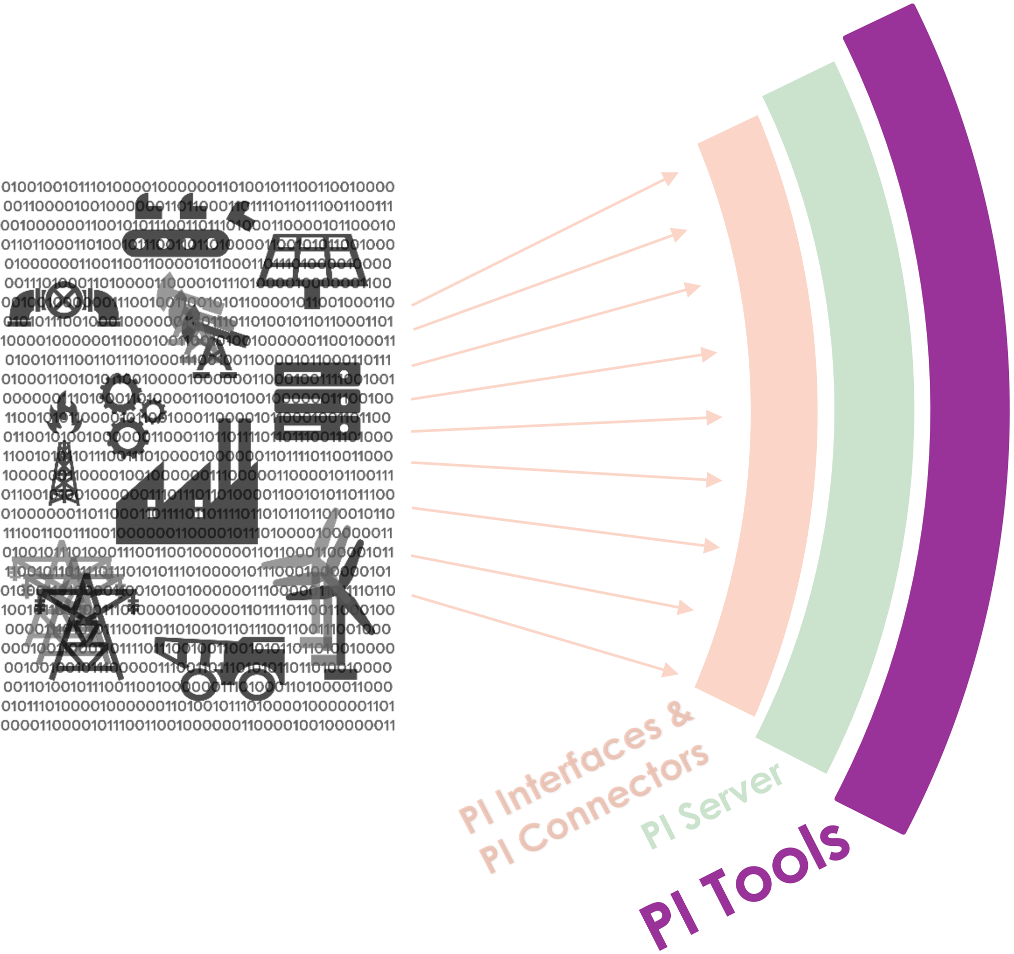


Stores high-frequency data.

Provides context to your data

Runs real-time analytics on your raw data

3. Deliver



Visualizes your data

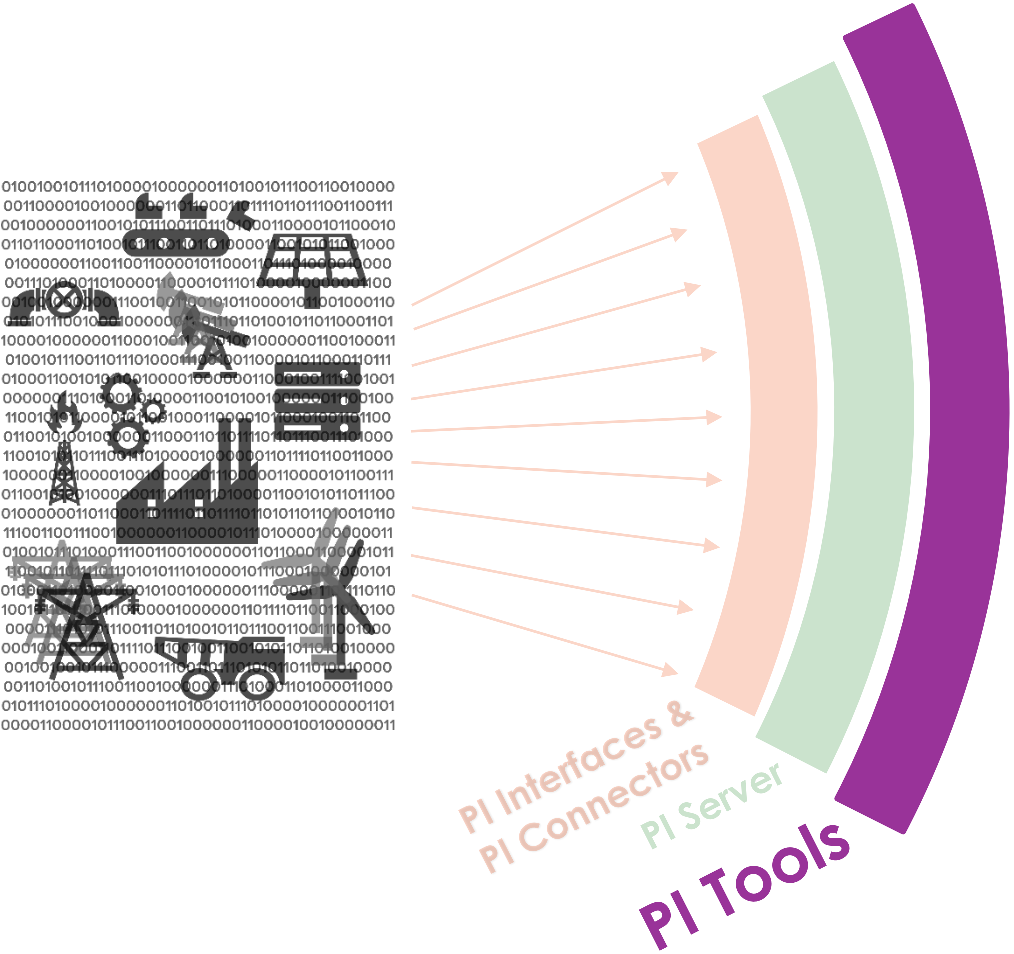
PI DataLink
Access data from Excel

PI ProcessBook
Visualize process in real time

PI Vision
Anywhere, Anytime

4. Deliver

SAP esri Microsoft TIBCO tableau



PI Integrators
Move data between business and geographic boundaries.

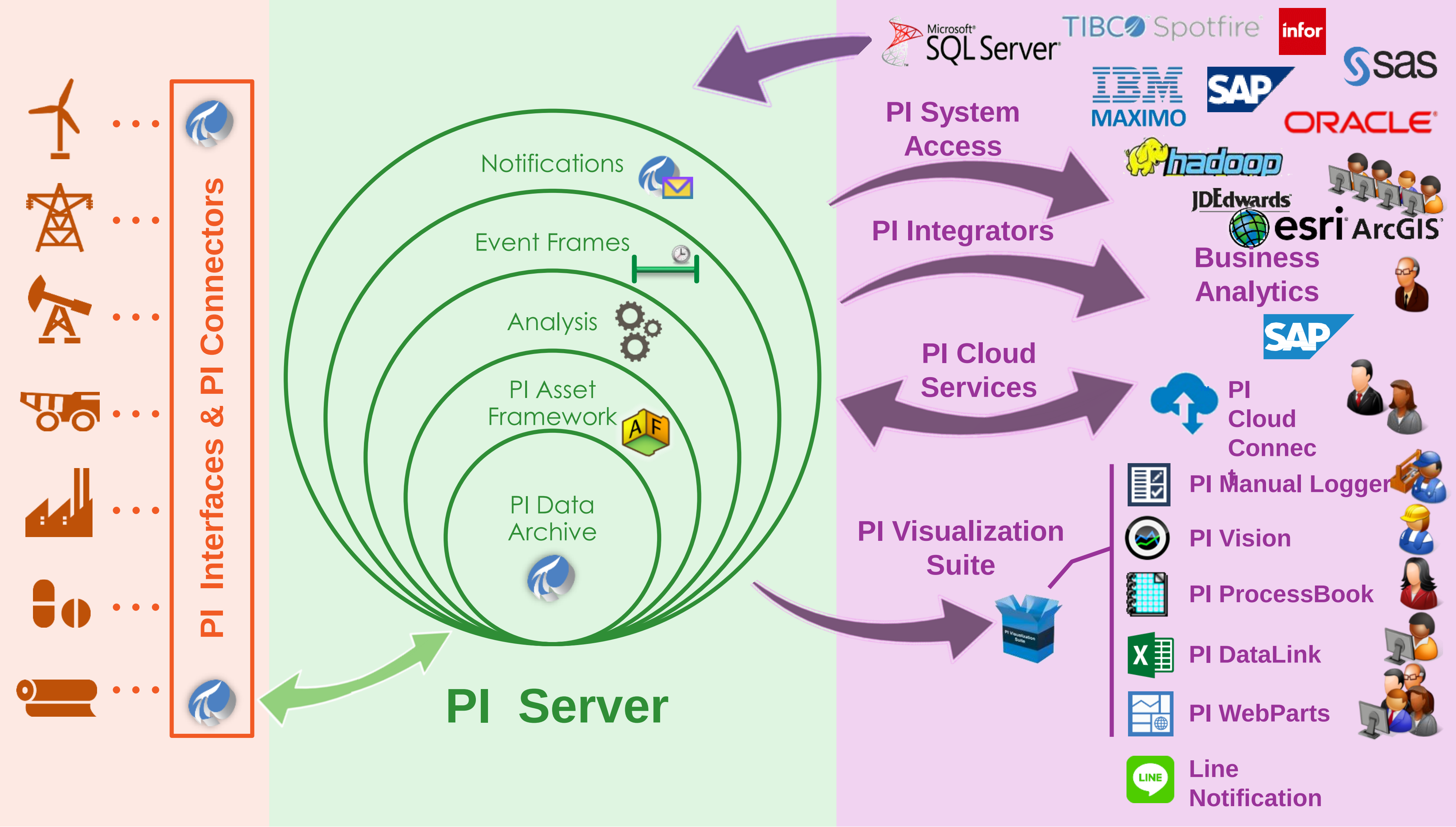
Shares your data

PI Data Access
</>
Programmatic access to build anything on PI

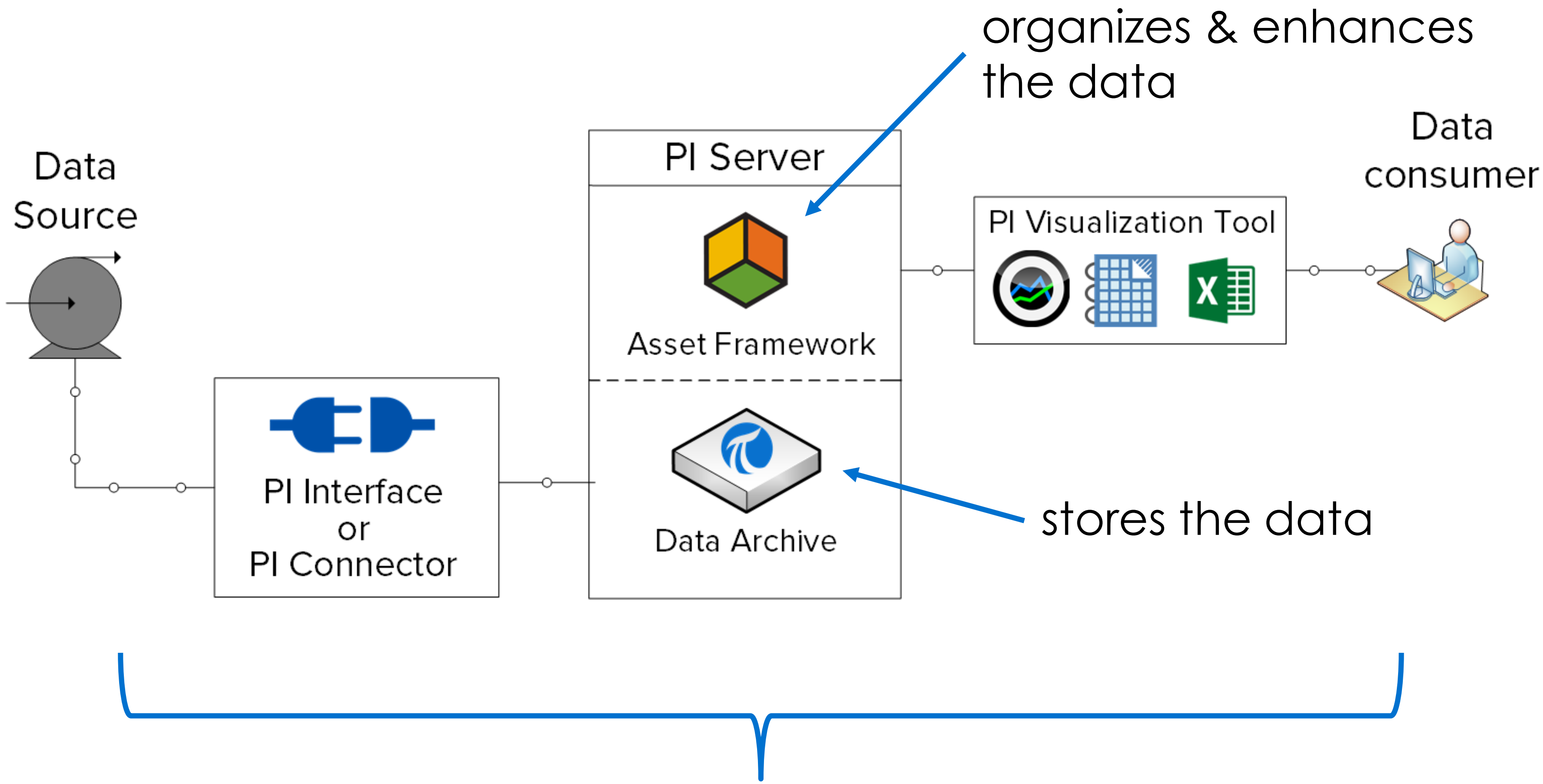
VisualKPI
www.transpara.com

Partner Applications

PI System Infrastructure

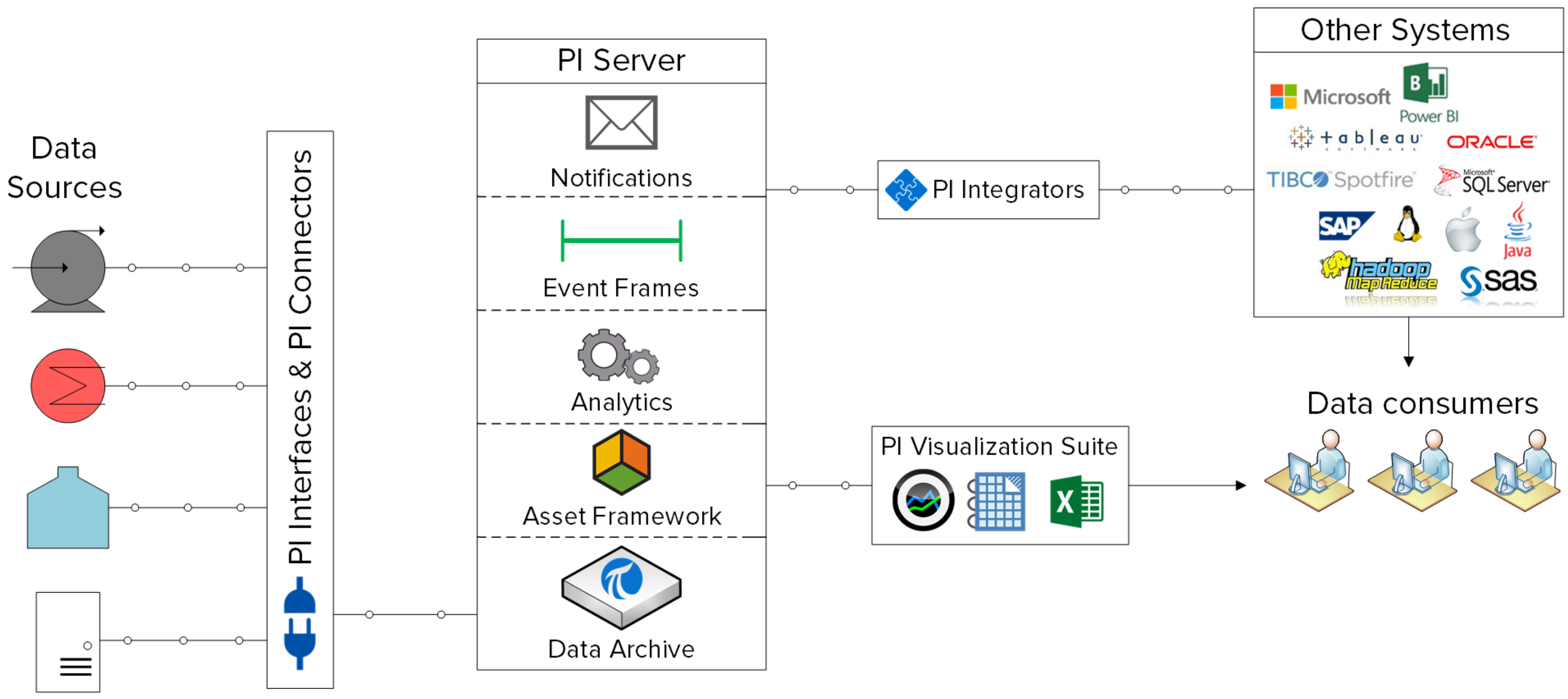


Basic components of a PI System

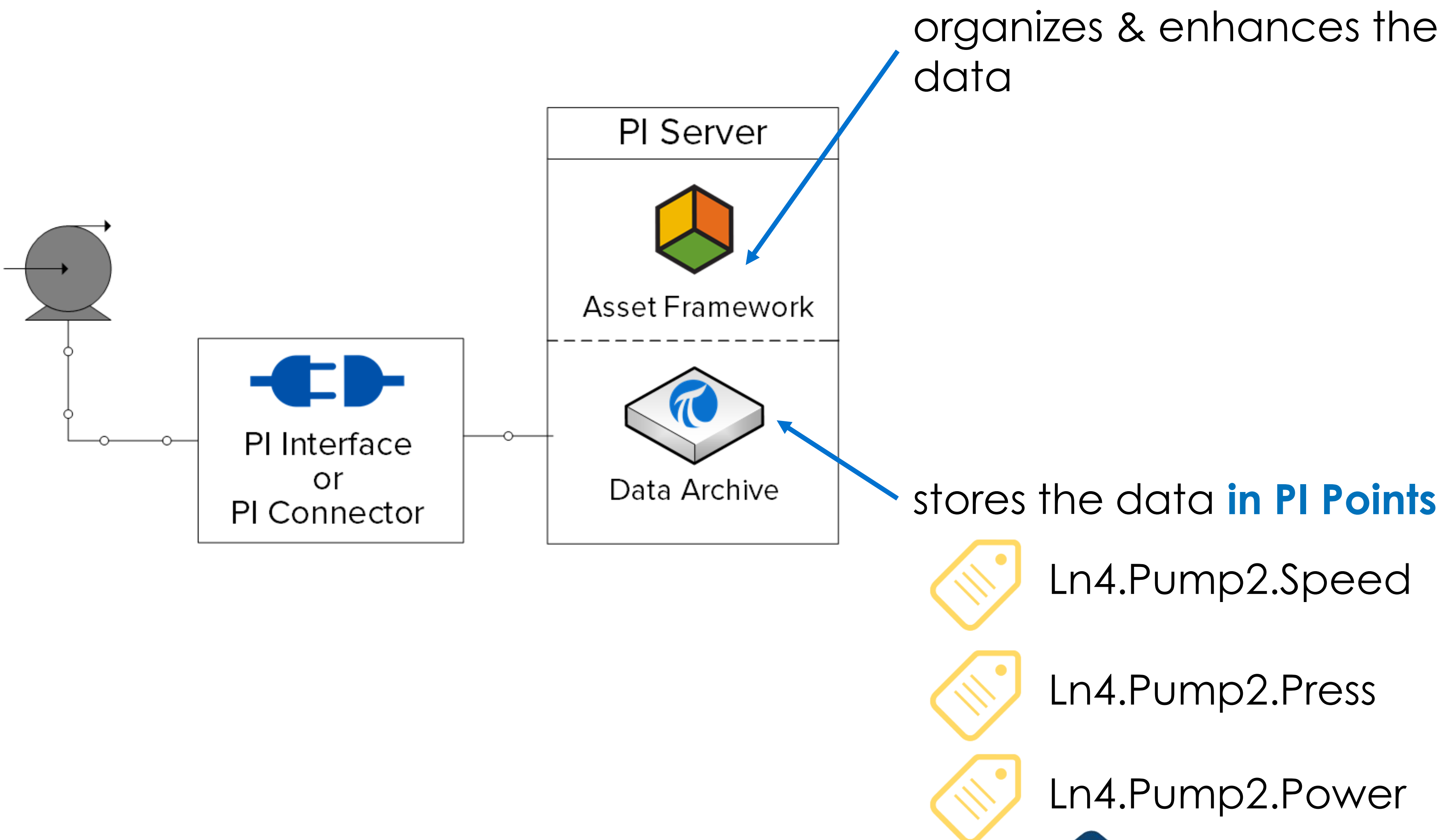


Software components!!

More complete PI System



What is a PI Point?

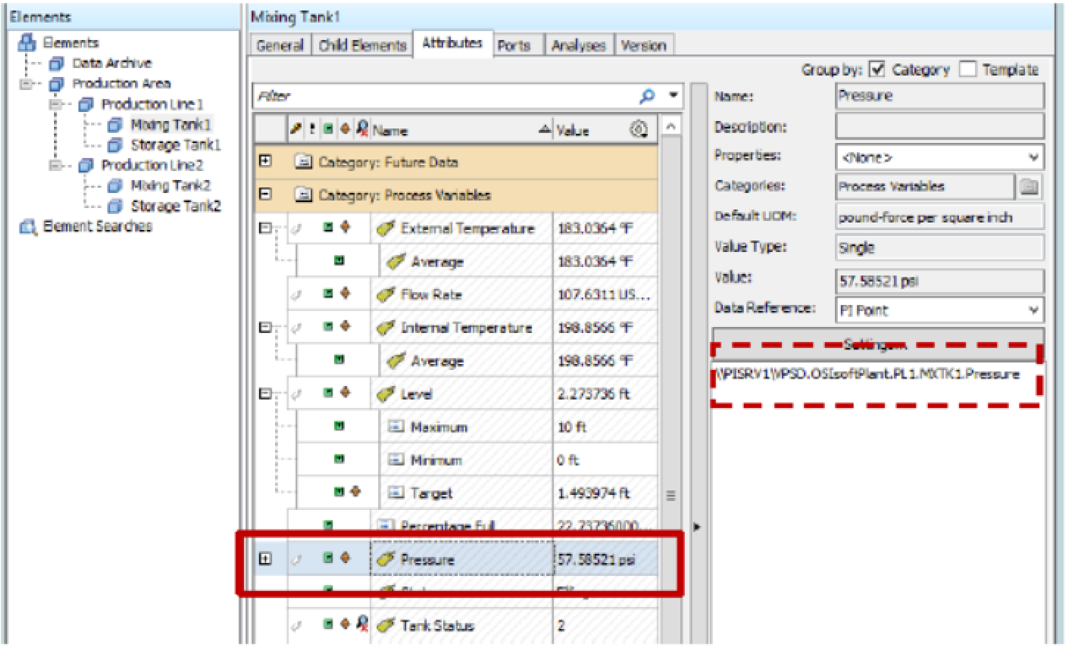


PI Tag and AF

PI Tags

VPSD.OSIsoftPlant.PL1.MXTK1.External Temperature
VPSD.OSIsoftPlant.PL1.MXTK1.Flow Rate
VPSD.OSIsoftPlant.PL1.MXTK1.Internal Temperature
VPSD.OSIsoftPlant.PL1.MXTK1.Level
VPSD.OSIsoftPlant.PL1.MXTK1.Level_Forecast
VPSD.OSIsoftPlant.PL1.MXTK1.Pressure
VPSD.OSIsoftPlant.PL1.MXTK1.TankStatus
VPSD.OSIsoftPlant.PL1.STTK1.External Temperature
VPSD.OSIsoftPlant.PL1.STTK1.Flow Rate
VPSD.OSIsoftPlant.PL1.STTK1.Internal Temperature
VPSD.OSIsoftPlant.PL1.STTK1.Level
VPSD.OSIsoftPlant.PL1.STTK1.Level_Forecast
VPSD.OSIsoftPlant.PL1.STTK1.Pressure
VPSD.OSIsoftPlant.PL1.STTK1.TankStatus
VPSD.OSIsoftPlant.PL2.MXTK2.External Temperature
VPSD.OSIsoftPlant.PL2.MXTK2.Flow Rate

AF Database



PI Point attributes

- **Name:** The name of the PI Point, which must be unique within the PI Data Archive.

- **Description**: A free text field attached to a PI Point, often used to enter a human friendly description of the PI Point.

- **Point Type:** This attributes defines the type of data that is stored in the PI Data Archive.

- **Point Source**: This attribute commonly specifies which PI Interface is collecting the data for the PI Point.

Point Type

Digital	• Discrete value (On/Off, Red/Black/Green)
Int16	• Integer value, 16 bits (0 to 32767, acc: 1/32767)
Int32	• Integer value, 32 bits (-2147450880 to 2147483647)
Float16	• Scaled Floating Point number, 16 bits (acc: 1/32767)
Float32	• Floating Point number, 32 bits (single precision)
Float64	• Floating Point number, 64 bits (double precision)
String	• Text value up to 976 characters
Blob	• Binary large object up to 976 bytes
Timestamps	• Any Time/Date in the range 1-Jan-1970 to 1-Jan-2038

Relative time expression syntax

Reference Time ± Offset

Abbreviation	Meaning
*	Now
t	Today at midnight
y	Yesterday at midnight
sun	Sunday at midnight
mon	Monday at midnight
tue	Tuesday at midnight
wed	Wednesday at midnight
thu	Thursday at midnight
fri	Friday at midnight
sat	Saturday at midnight
YYYY	The current day and month of year YYYY at midnight
M-D or M/D	The Dth day of the Mth month of the current year at midnight
DD	The DDth day of the current month and year at midnight

Abbreviation	Time-unit
s	second
m	minute
h	hour
d	day
w	week
mo	month
y	year

Relative time Examples

Expression	Meaning
*-1h	One hour ago
t+8h	08:00:00 (8:00 a.m.) today
y-8h	16:00:00 (4:00 p.m.) the day before yesterday
mon+14.5h	14:50:00 (2:30 p.m.) last Monday
sat-1m	23:59:00 (11:59 p.m.) last Friday

Future Data

Future timestamp

Time range of January, 1970 through January, 2038

Input	Meaning
*+1h	An hour from now
t+3d	Three days from today at midnight
Y+1y	A year from yesterday

PI System time expression rules

1 Only include a single time offset

BAD: *+1d+4h

2 In offsets, only use fractions for hours, minutes and seconds

GOOD: *-1.5h

BAD: *-1.5d

3 For fixed timestamps:

- If time is not specified, default is midnight (00:00:00)
- If day is not specified, current day
- If month is not specified, current month
- If year is not specified, current year

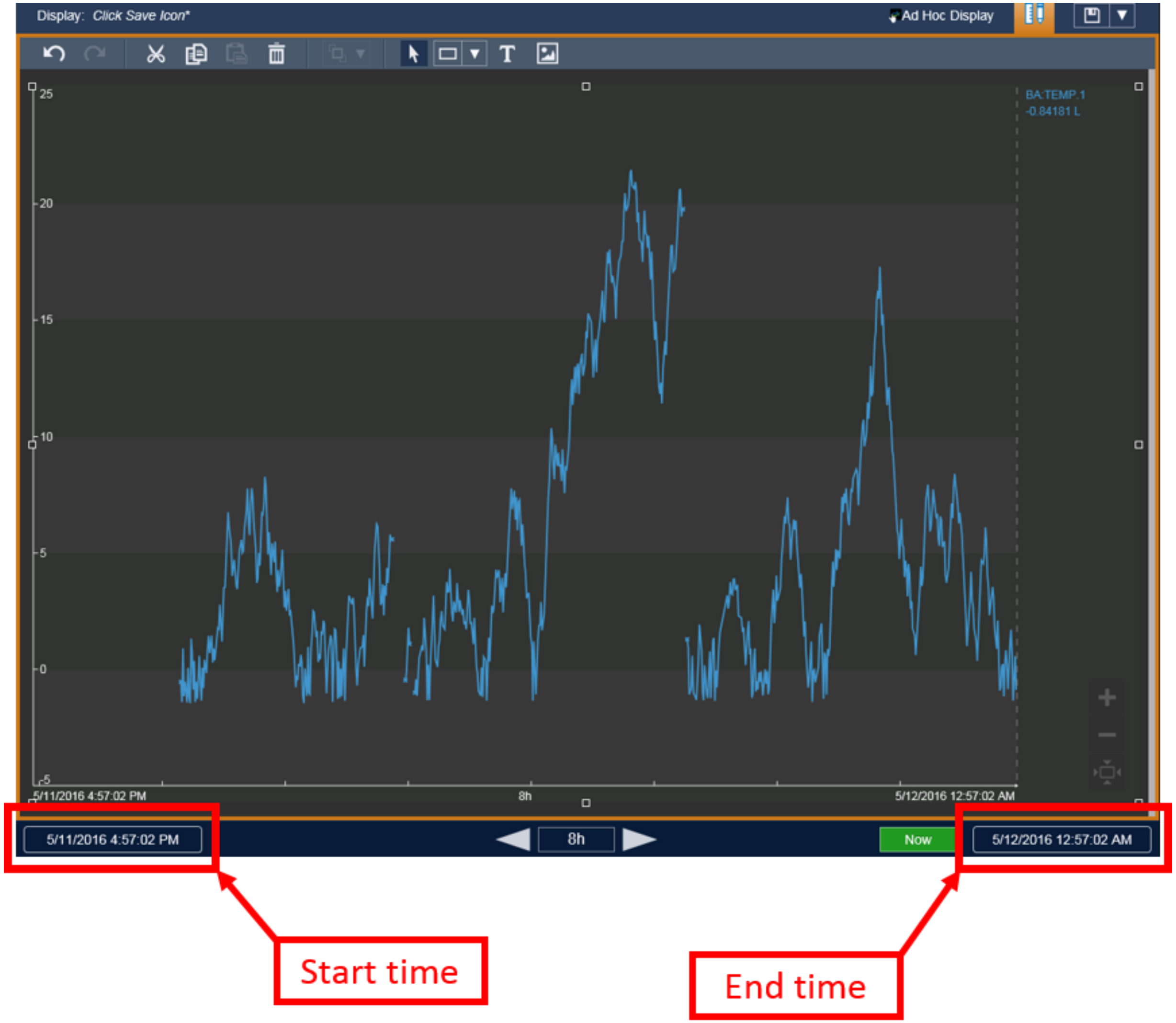
Directed Activity: PI System times

Expression	Meaning
* - 30m	
y + 8h	
T	
Y	
Thu	
Tuesday – 2d	
18	
y-2y	

Directed Activity: PI System times

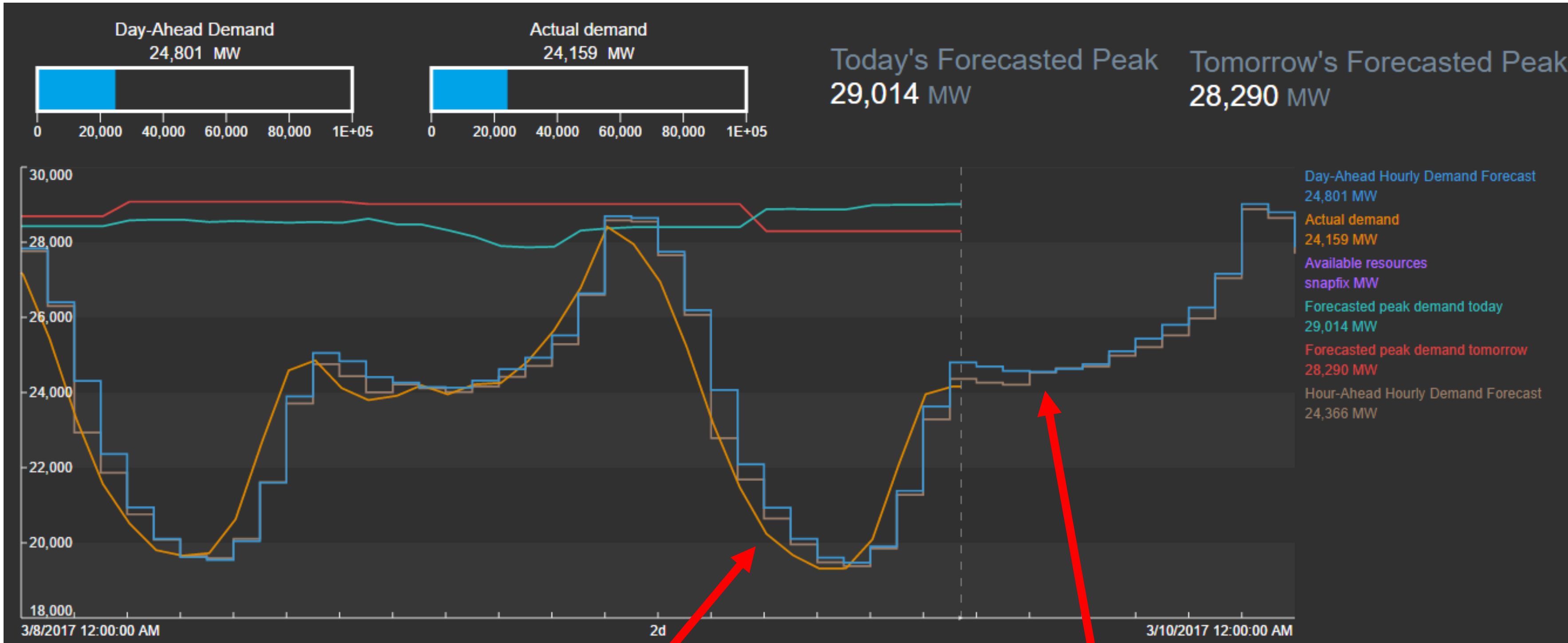
Expression	Meaning
	Today at 6:00 AM
	Monday at 6:30 AM
	12 hours ago
	The first day this month
	The end of this week (this Friday)
	7:00 am yesterday
	15 minutes ago

Directed Activity: PI System times



Use your knowledge of PI System time abbreviations to trend data on your "Reactor 1 temperature" display

Future Data



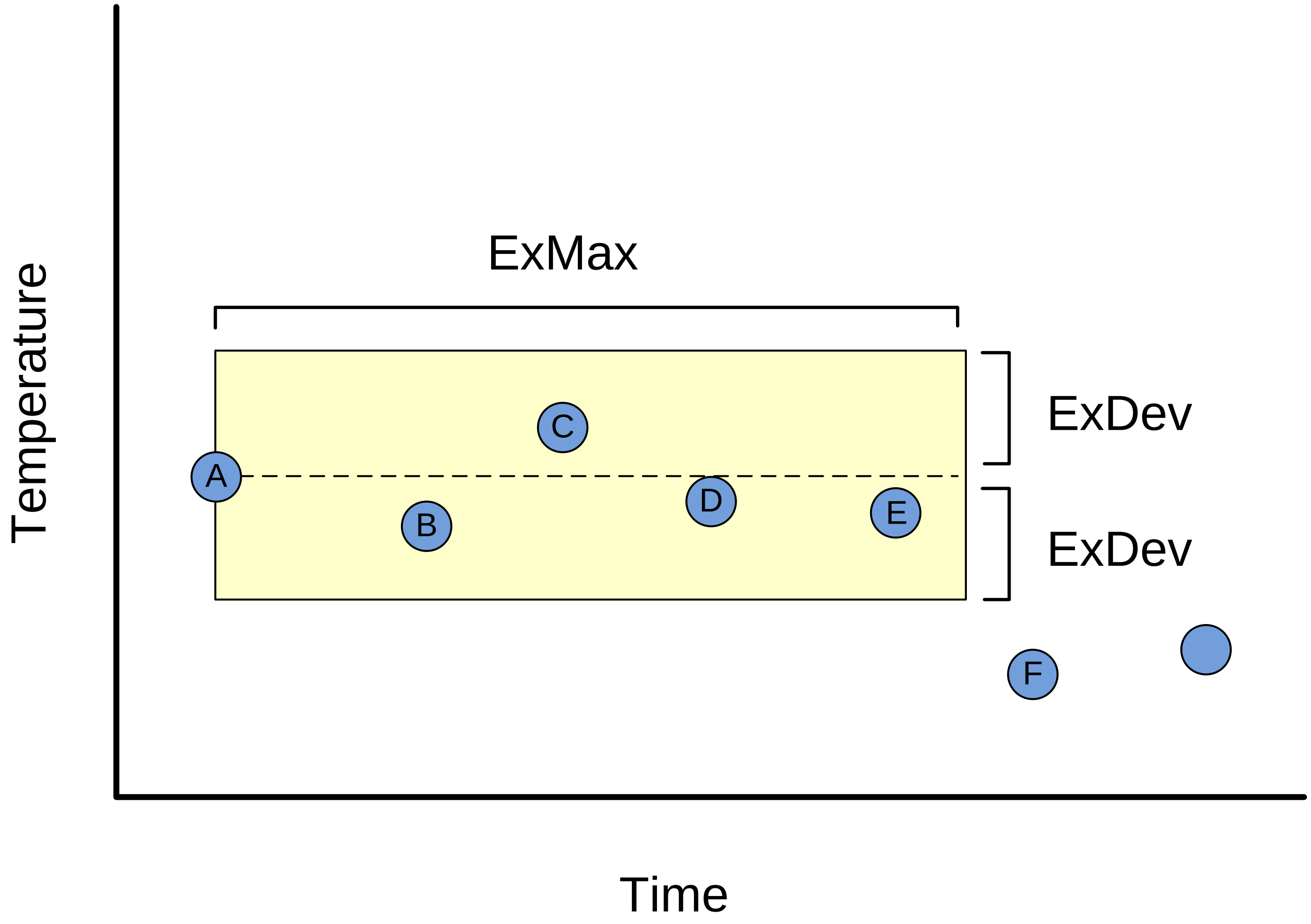
Actual Demand is a **real-time data** PI Point

Day ahead demand is a **future data** PI Point

<u>Once a PI Point is created as real-time or future, it can't be changed!!</u>

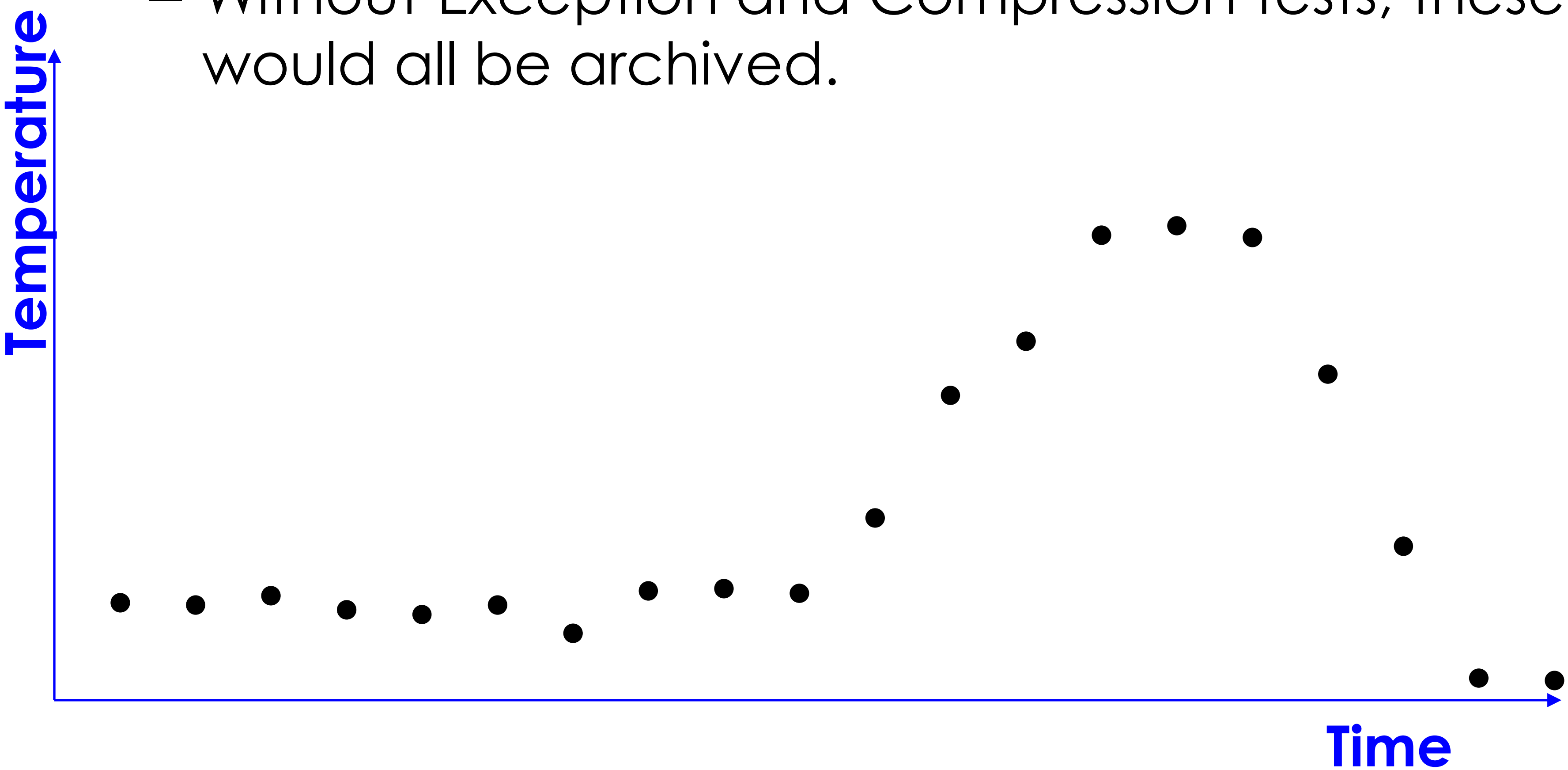
Solo Exercise: Filtering data

Exception: Filtering Noise

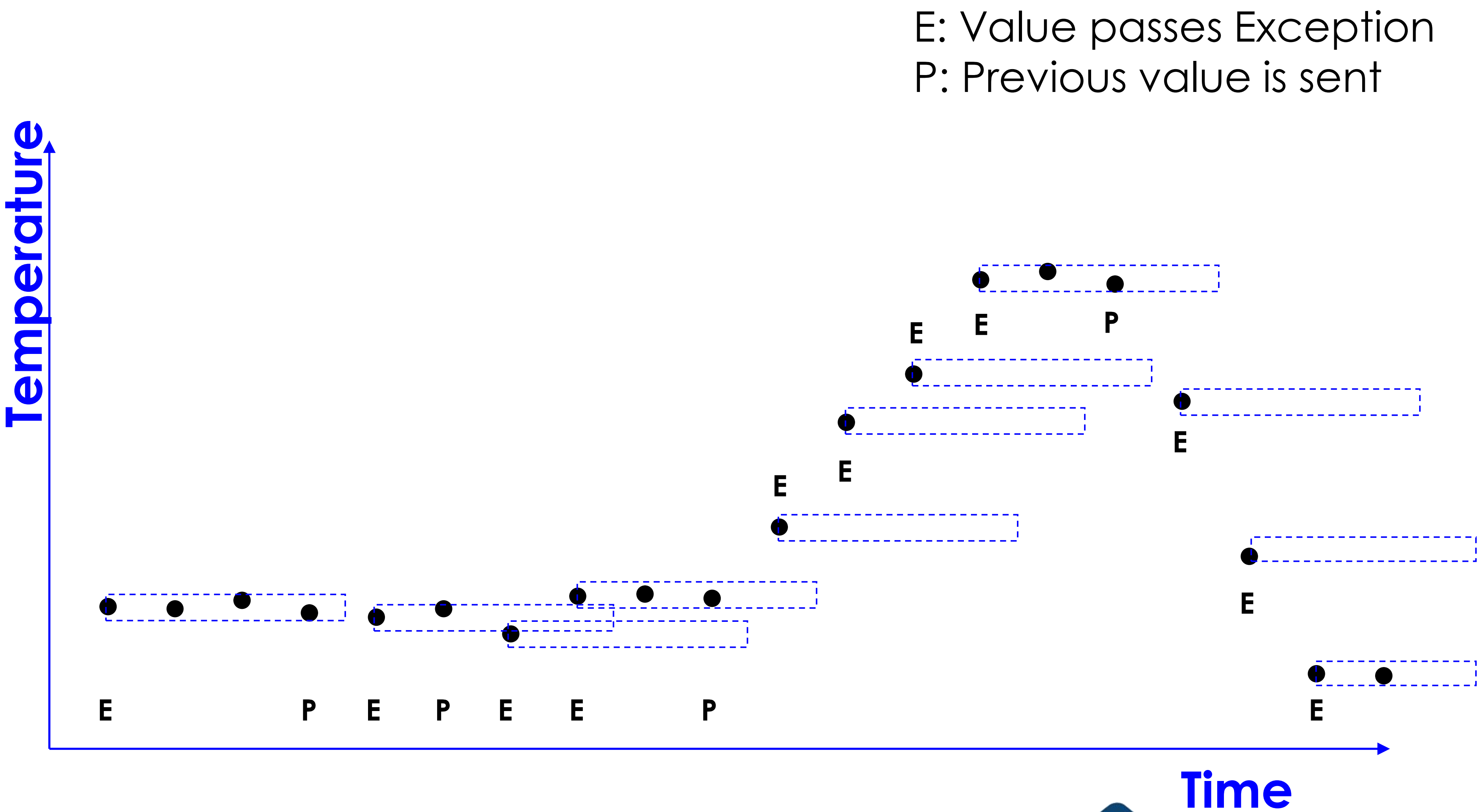


Raw Data - Example

- Raw values scanned on the data source.
  - Without Exception and Compression tests, these would all be archived.

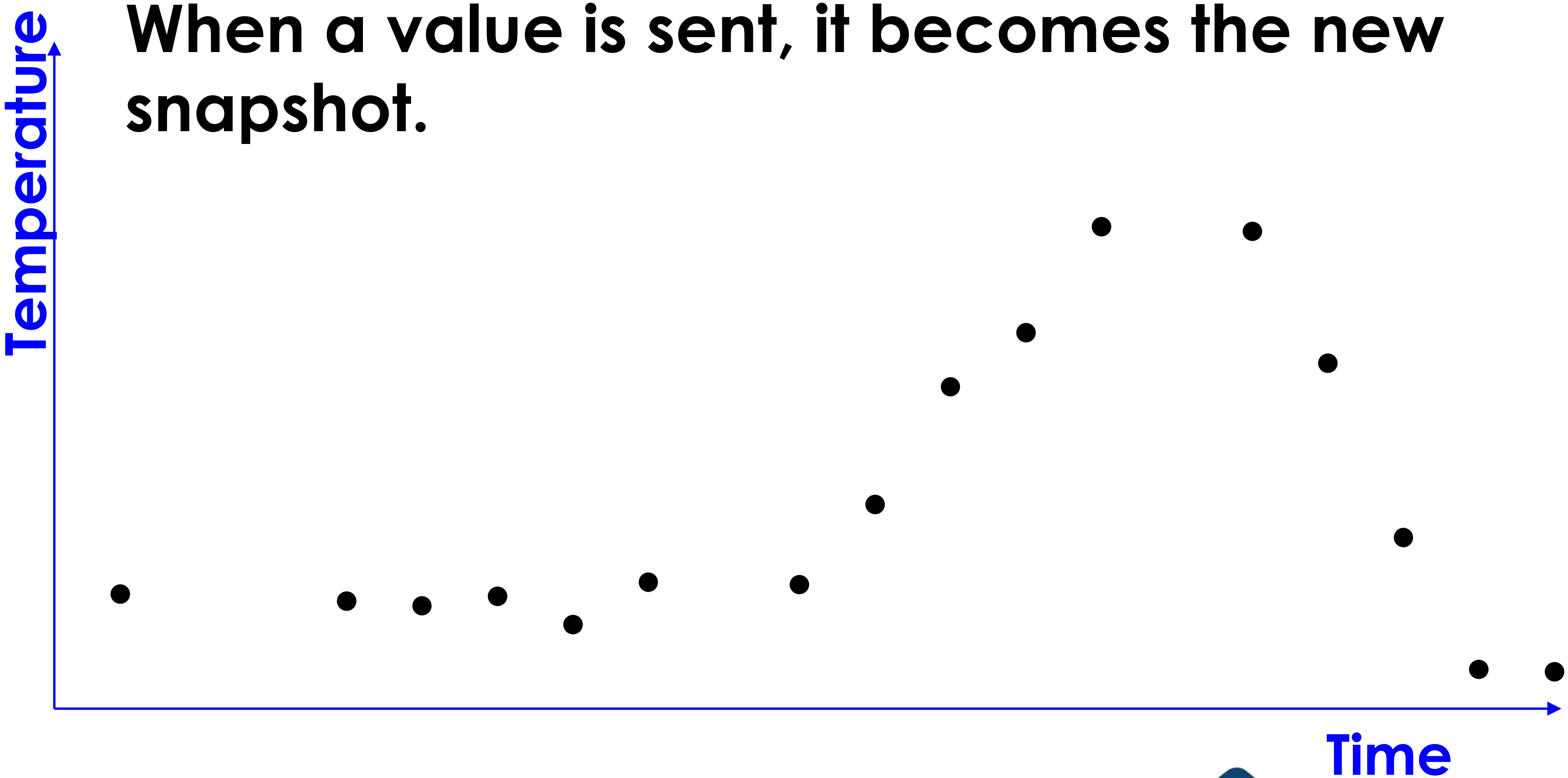


Visualizing the Exception Test

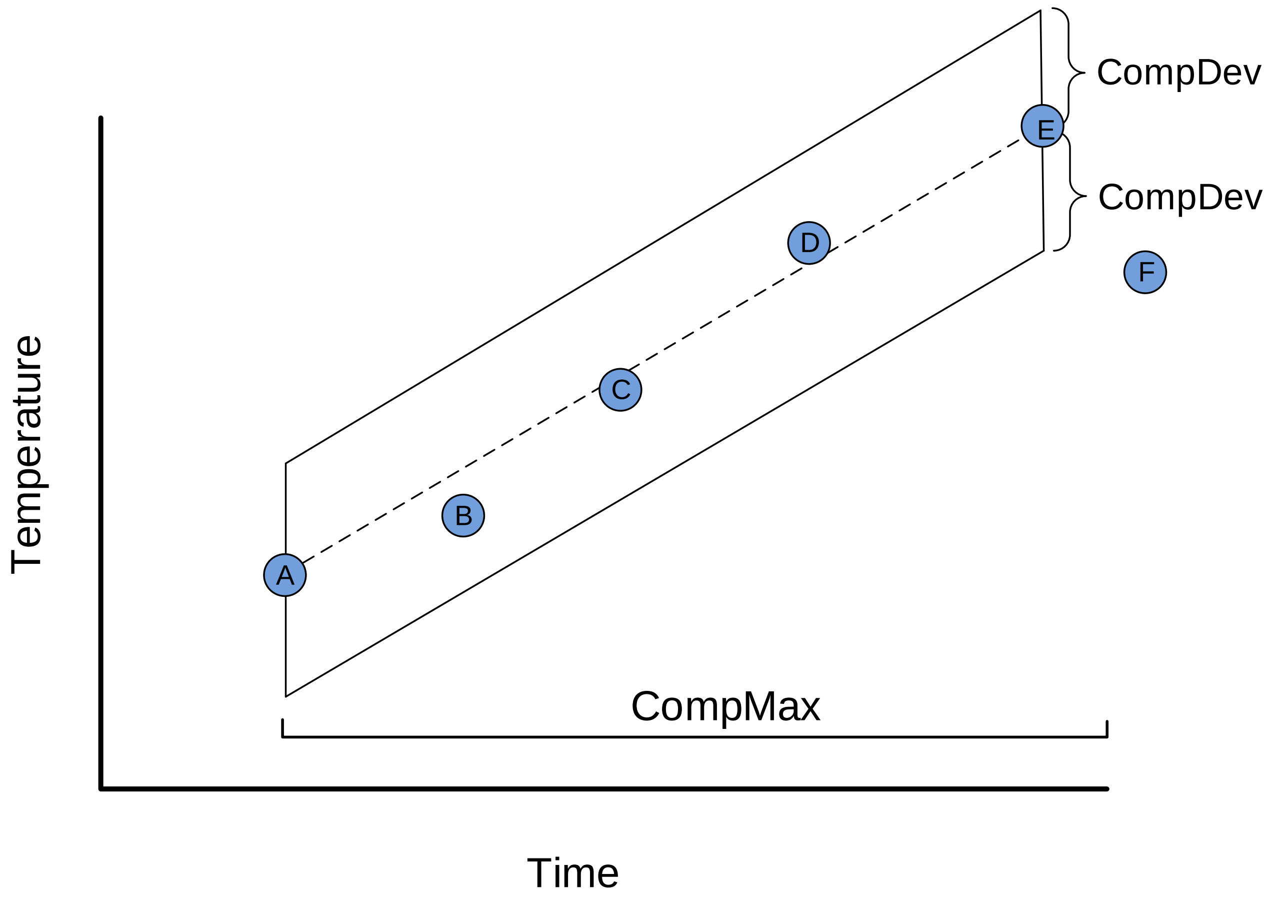


Exception Test - Results

Successive values sent to the PI Server. When a value is sent, it becomes the new snapshot.

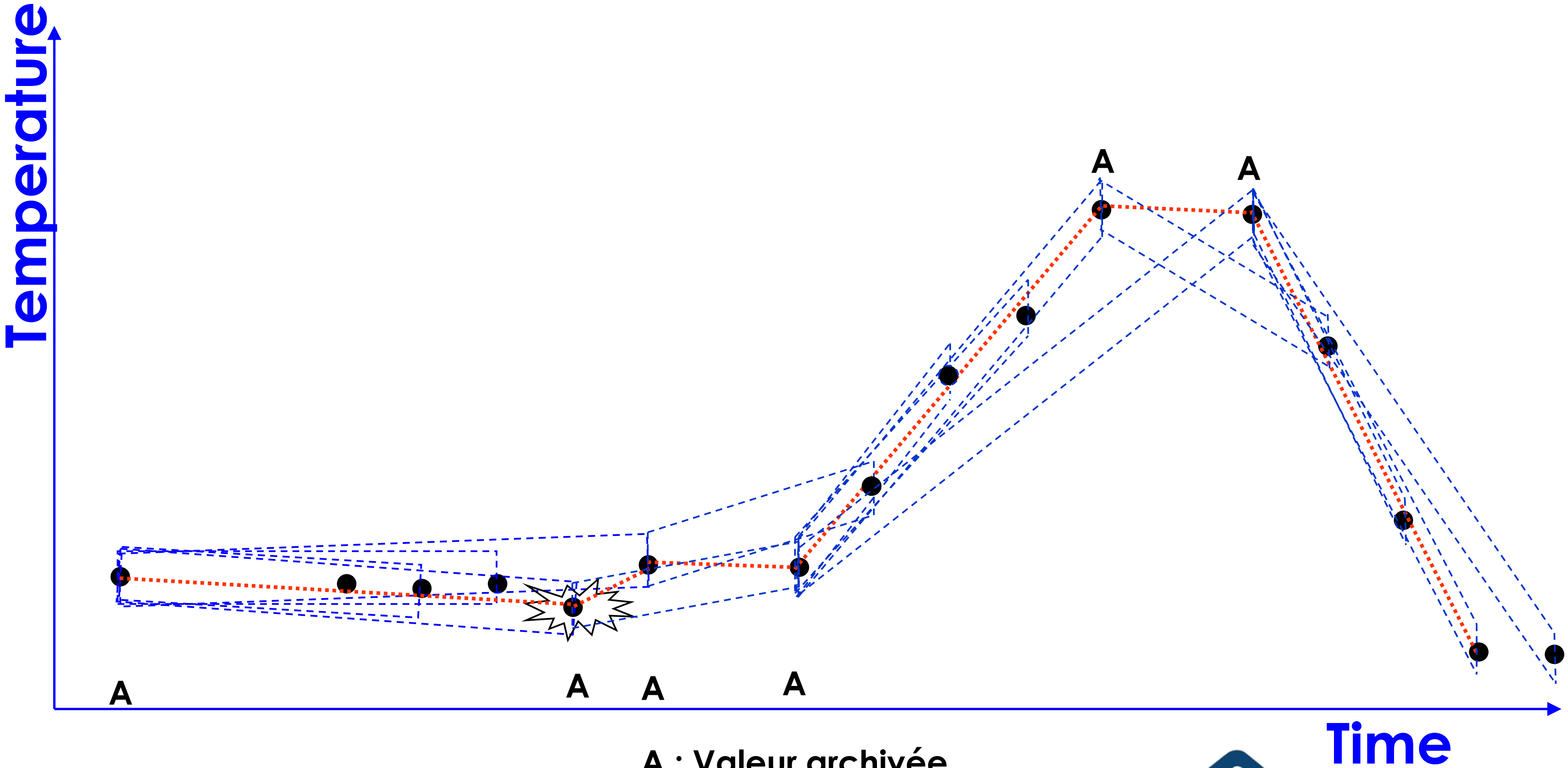


Storing Only Meaningful Data



Visualizing the Compression Test

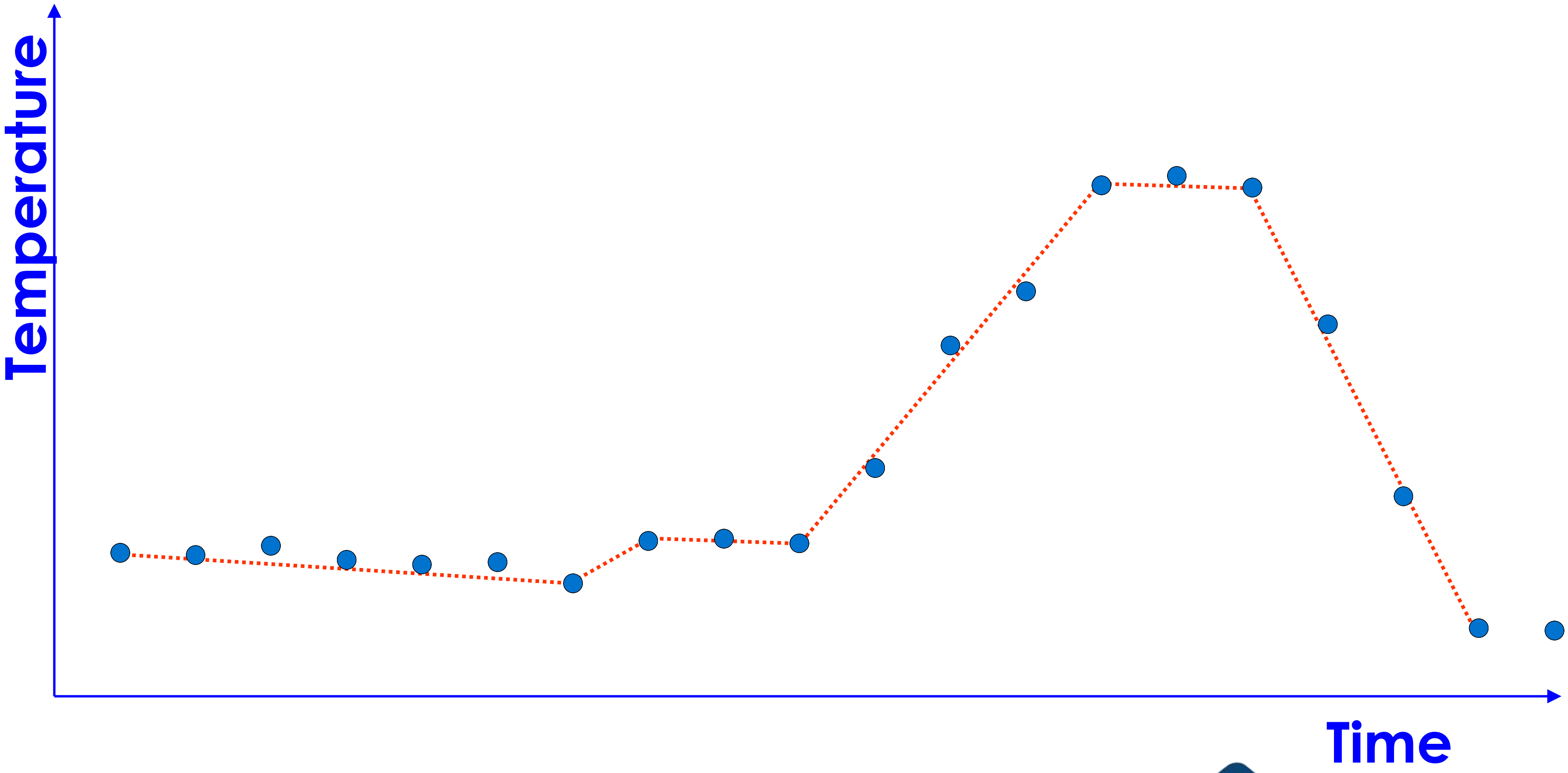
A: Value is Archived



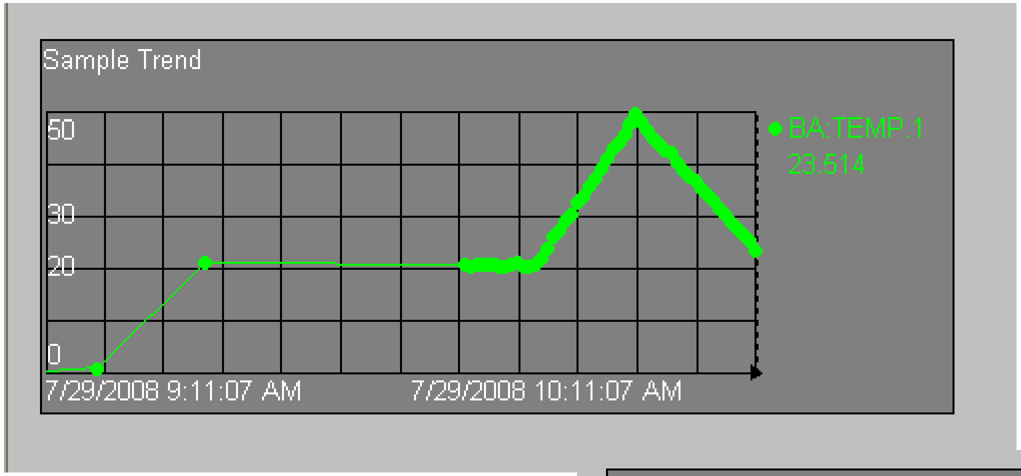
A : Valeur archivée

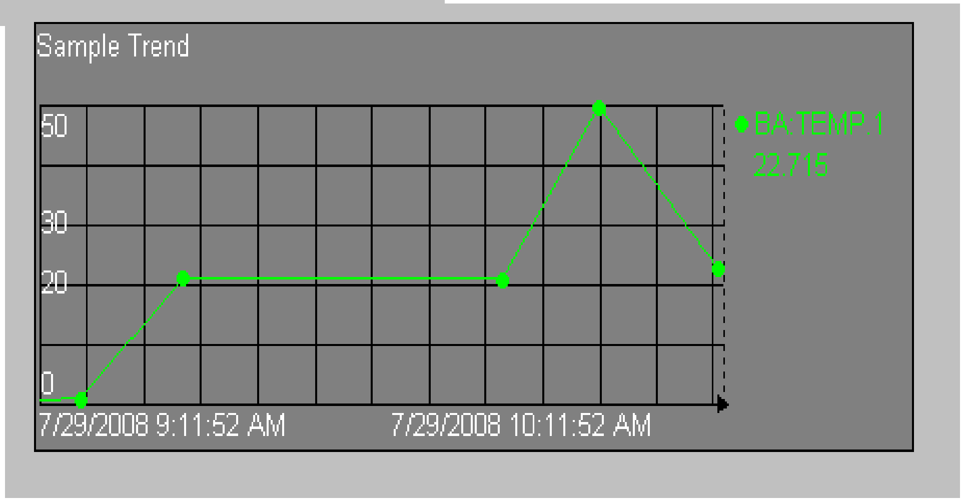
Cumulative Results

Raw values scanned.



Effect of compression on displays

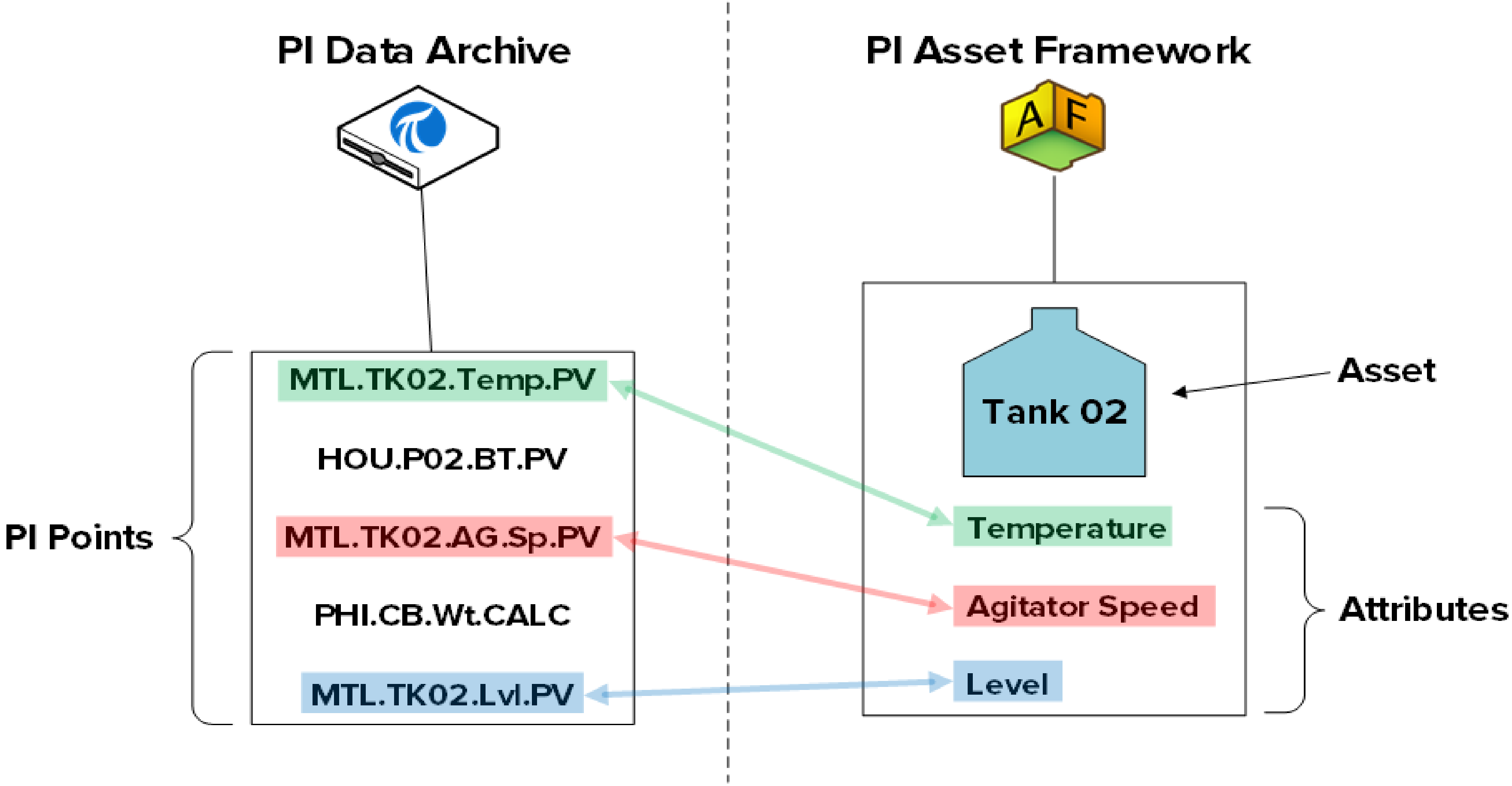




PI Asset Framework Management

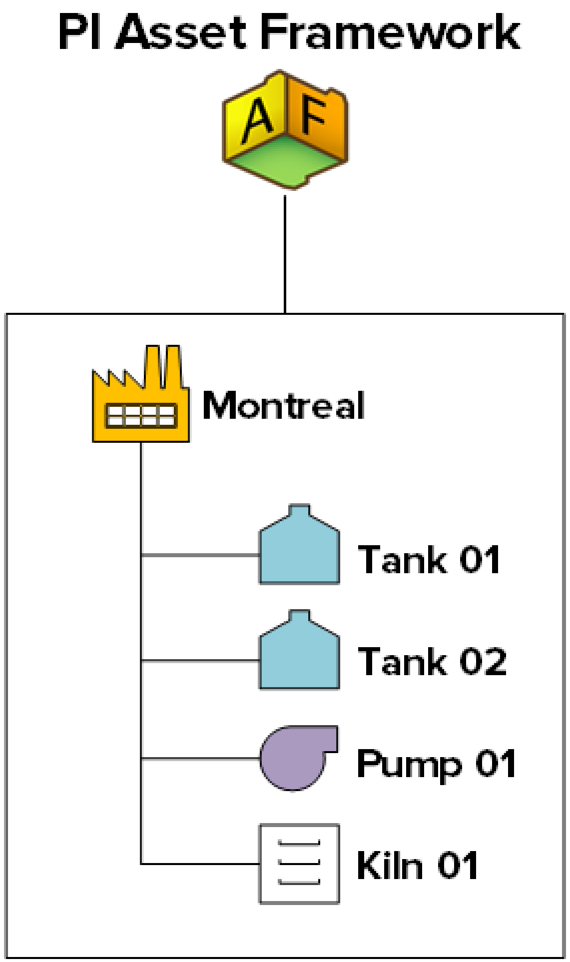
How data is organized

The data is organized under **assets**



How data is organized

Assets are organized in a **hierarchy**



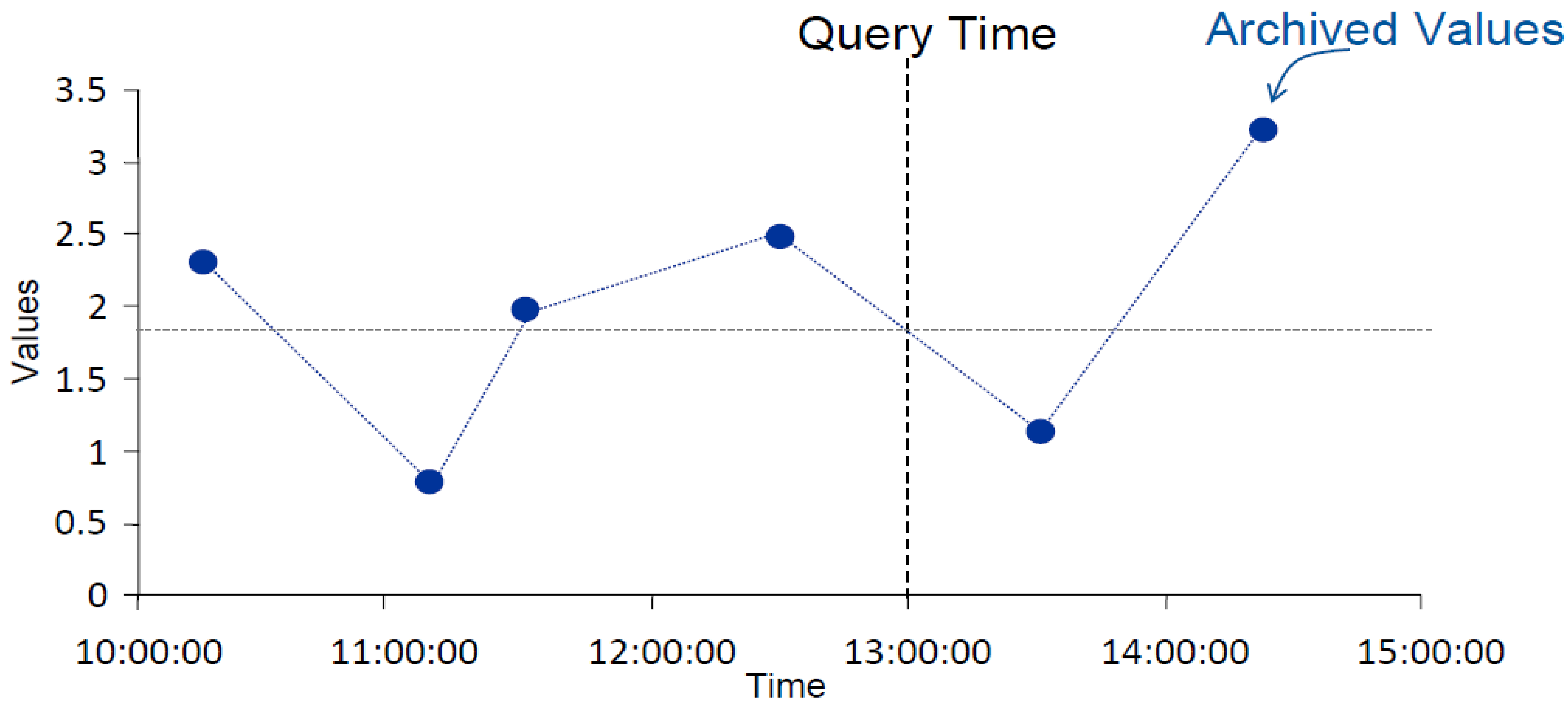
1. Human friendly
2. Can easily calculate things about Montreal plant
3. Can easily compare similar assets (e.g. Tanks)

PI Datalink and PI Vision

PI DataLink

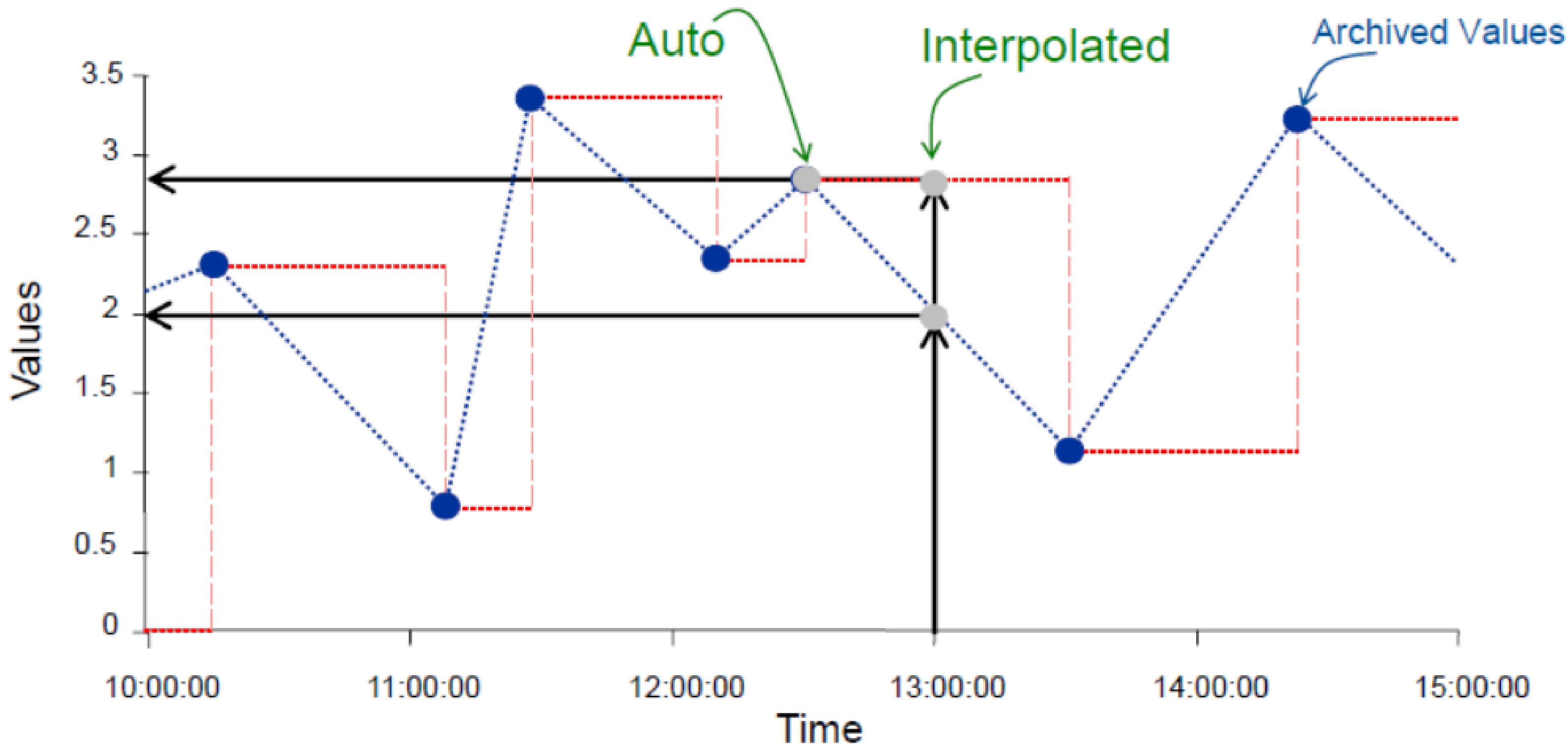
PI Vision

Retrieval Mode



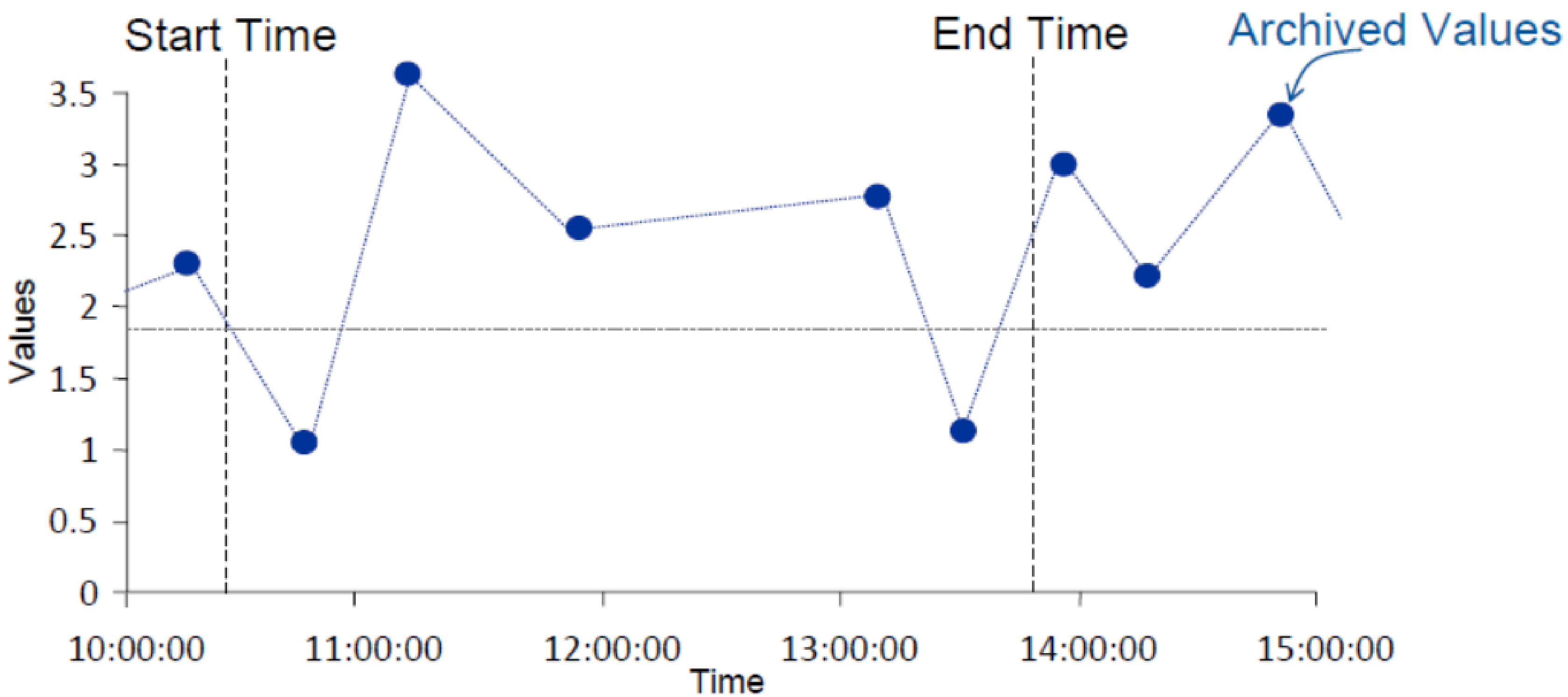
Retrieval Mode	Timestamp	Value
Interpolated	13:00:00	1.8
Previous	12:30:00	2.5
Next	13:30:00	1
Exact	No events found.	No events found.

Retrieval Mode-interpolated versus



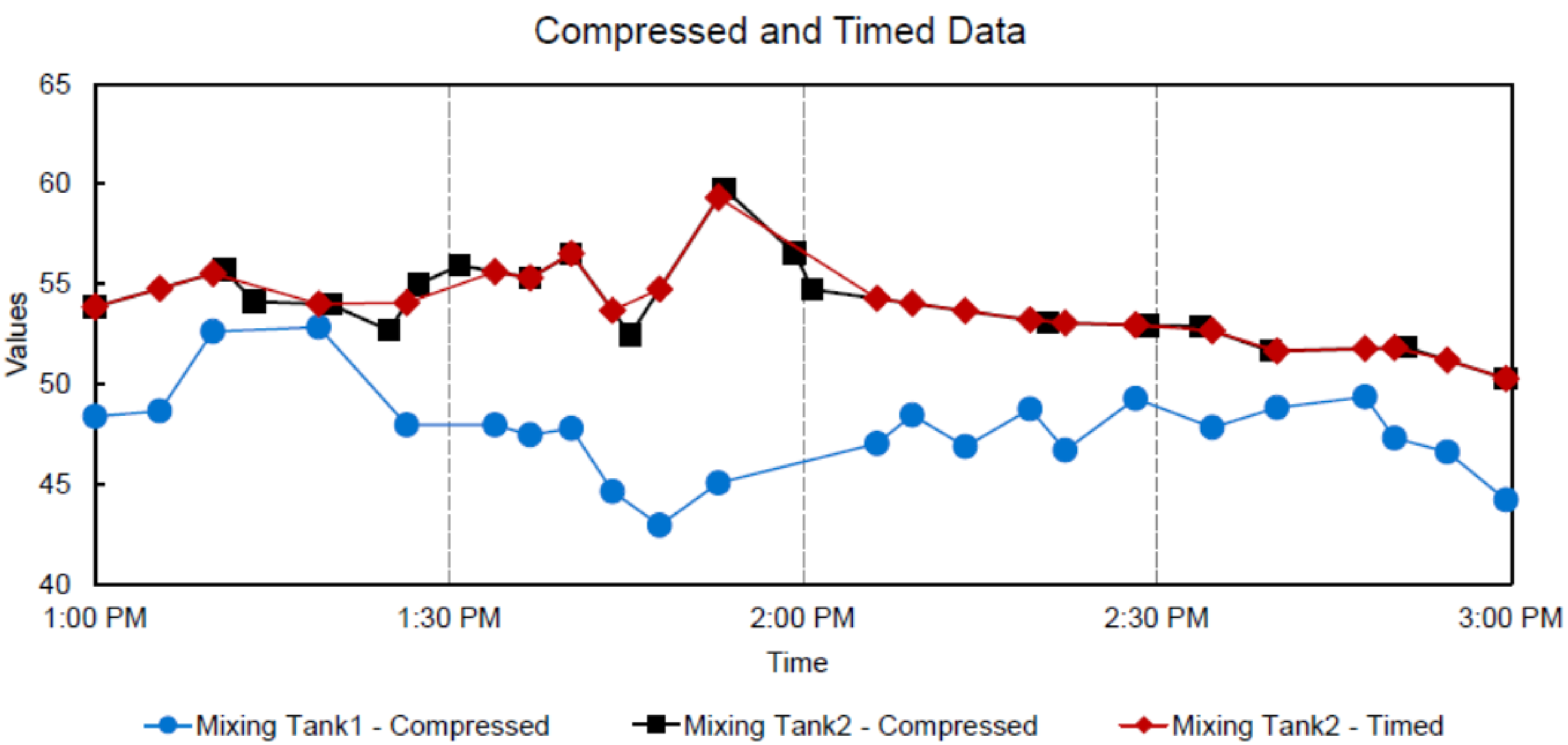
	Auto	Interpolated
Step OFF (Rate tag)	13:00:00 - 2	13:00:00 - 2
Step ON	12:30:00 – 2.8	13:00:00 – 2.8

ComPressed Versus Sampled Value



Boundary Type	Number of Values Retrieved
Inside	5
Outside	7
Interpolated	7

Multiple Value Boundary Types



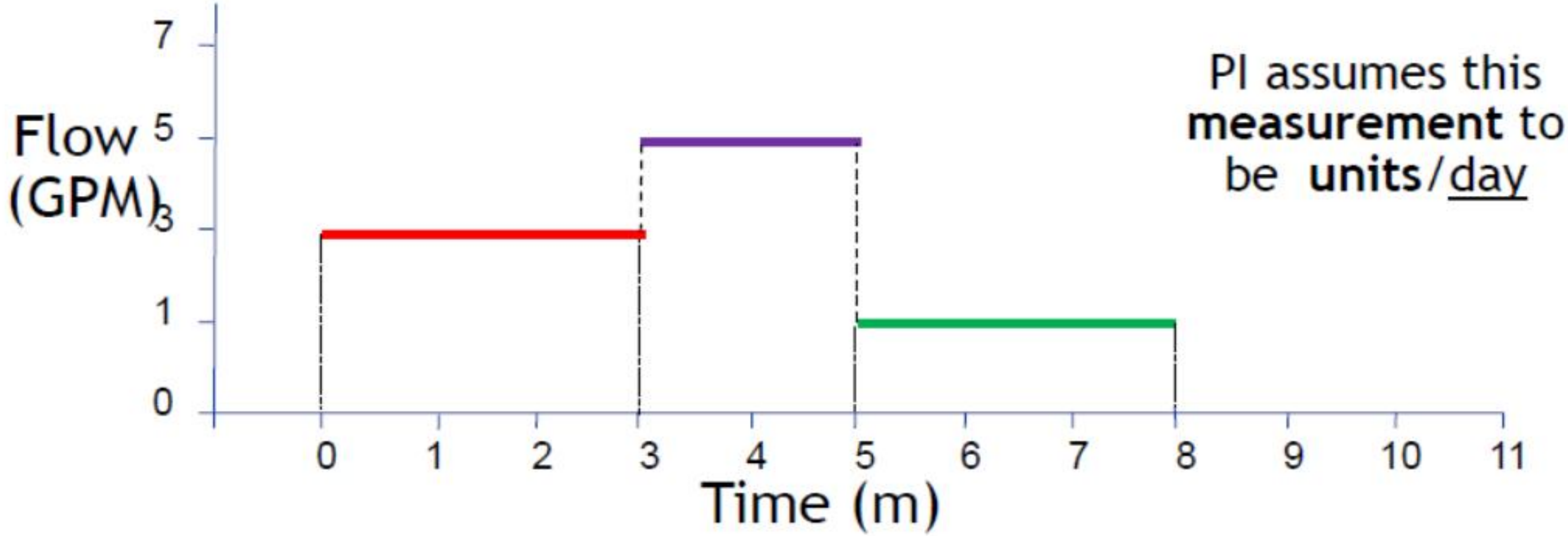
Compressed versus Timed Data

Rate Tag Engineering Units	Assumption of The PI Data Archive	Conversion Factor
Units / Day	Units / Day	1
Units / Hour	Units / Day	24
Units / Minute	Units / Day	1440
Units / Second	Units / Day	86400

SOLUBLE

Calculated Data

- Conversion Factor and Totalizations…

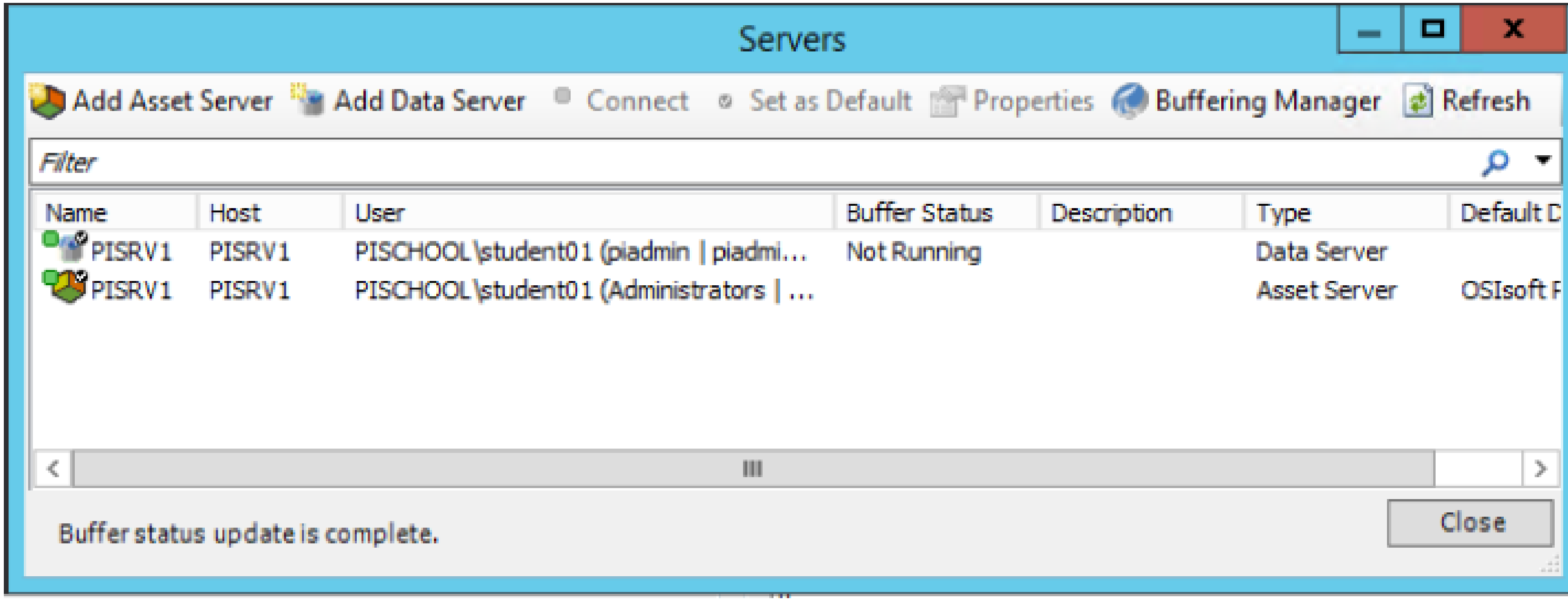


Normal total = (3 gpm x 3m) + (5 gpm x 2m) + (1 gpm x 3m) = 22 gallons

PI total = ((3 **gallons per** day x 3m x 1day/1440m) + (5 **gal**lons per day x 2m x 1day/1440m) + (1 **gallon per** day x 3m x 1day/1440m)) * ***1440m/1d*** = 22 gallons

The total computed by PI must be multiplied by a factor of 1440

Connecting to the PI System in PI DataLink



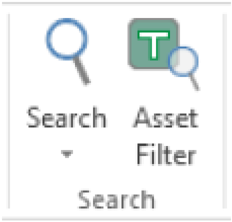
denotes a Data Archive

denotes a AF Server

Finding Data Items using PI DataLink Search

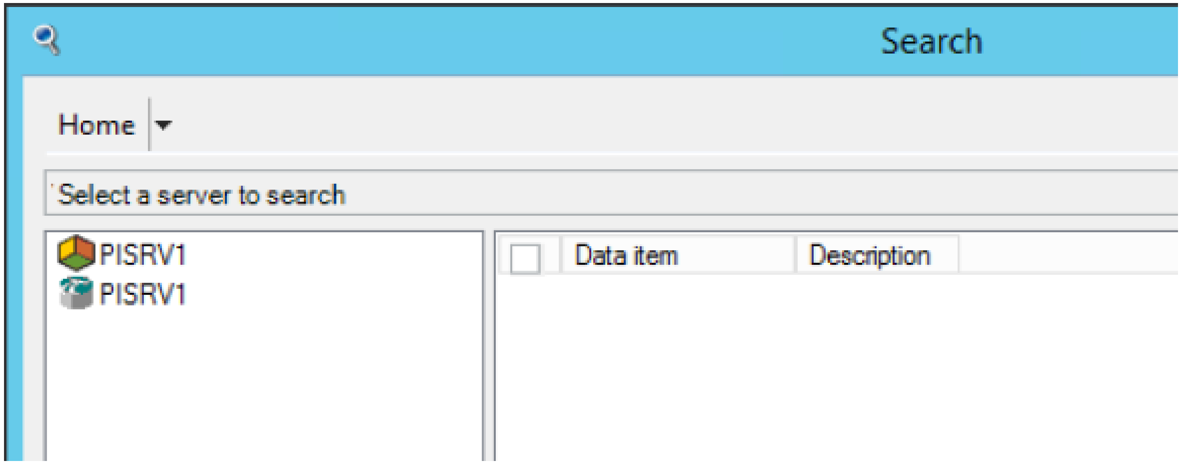
The search functionality has been updated and enhanced starting in PI DataLink and offers you two ways to search for data items:

1. Search tool
2. Asset Filter Search function (this will be covered in a later section)

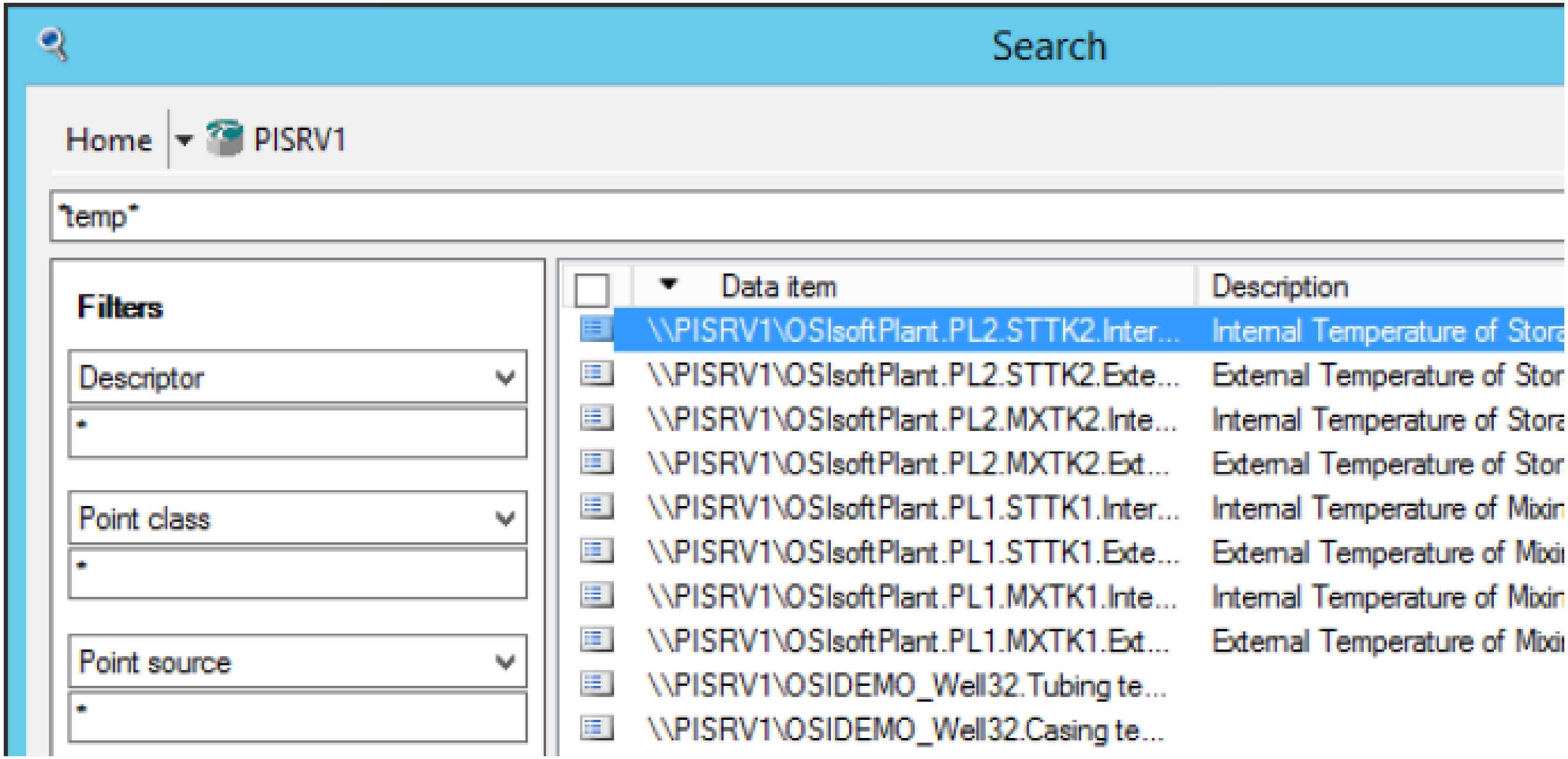


Search for Data Items

Upon first use, the tool starts at the Home node, which shows all the Data Archives and AF servers listed in Connection Manager. You must limit the search to a single Data Archive or single AF server, and can limit the search further to a single database on a AF server, and then to specific elements and parent attributes.



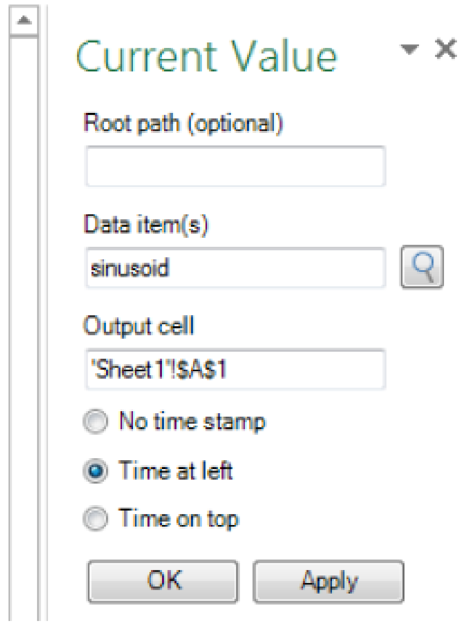
PI Point Name



When your n scope is a AF Server or Database, the top search field is applied to AF Attribute name as well as the name, description and categories of the parent element.

Current Value

- **Output cell**
  - Any data currently in this cell will be replaced.



The result of this query will be the most current value and timestamp of the data item specified.

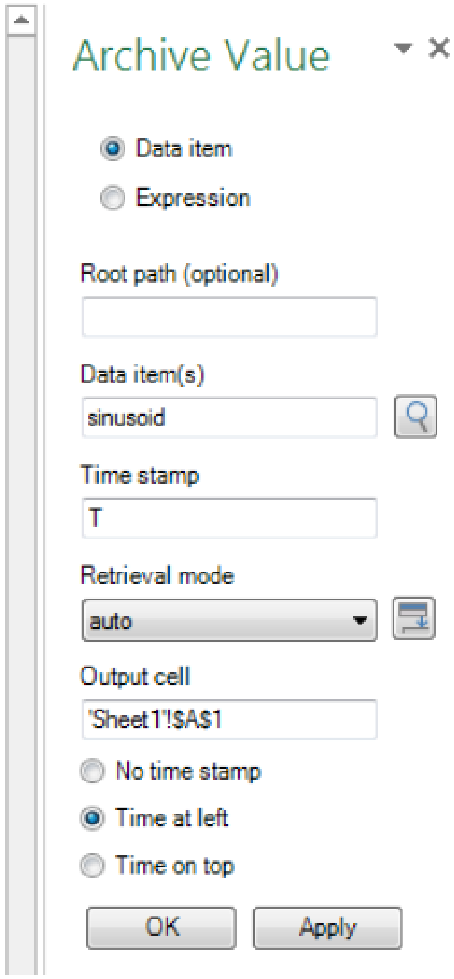
	A	B
1	02-May-14 10:44:01	19.222

Archive Value

The other PI DataLink function that returns a single value is the Archive Value. This function retrieves an archived value at a specific timestamp.

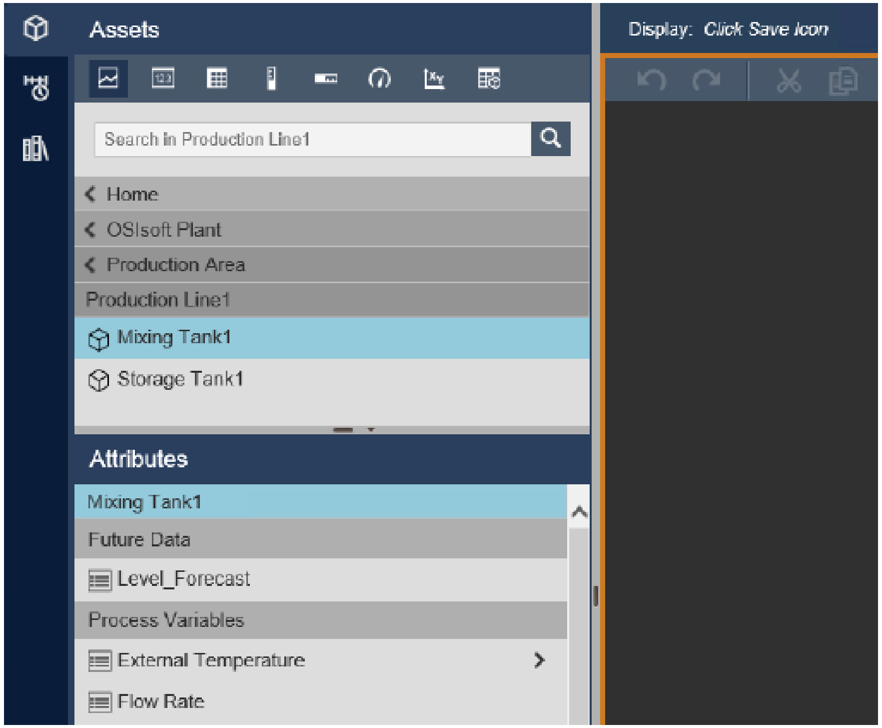
You must specify:

- Data item(s)
  - Can specify 1 or more
- Time stamp
  - Excel Time Format
  - PI Time Format
- Retrieval Mode
  - Several options, default is Auto.



PI Vision

To create a new display, click New Display  and from the top level AF Server, drill down by clicking on the black arrows to find assets in the plant. Notice the hierarchy of assets is displayed above. Once you have the asset of your interest, notice the Attributes list populates.



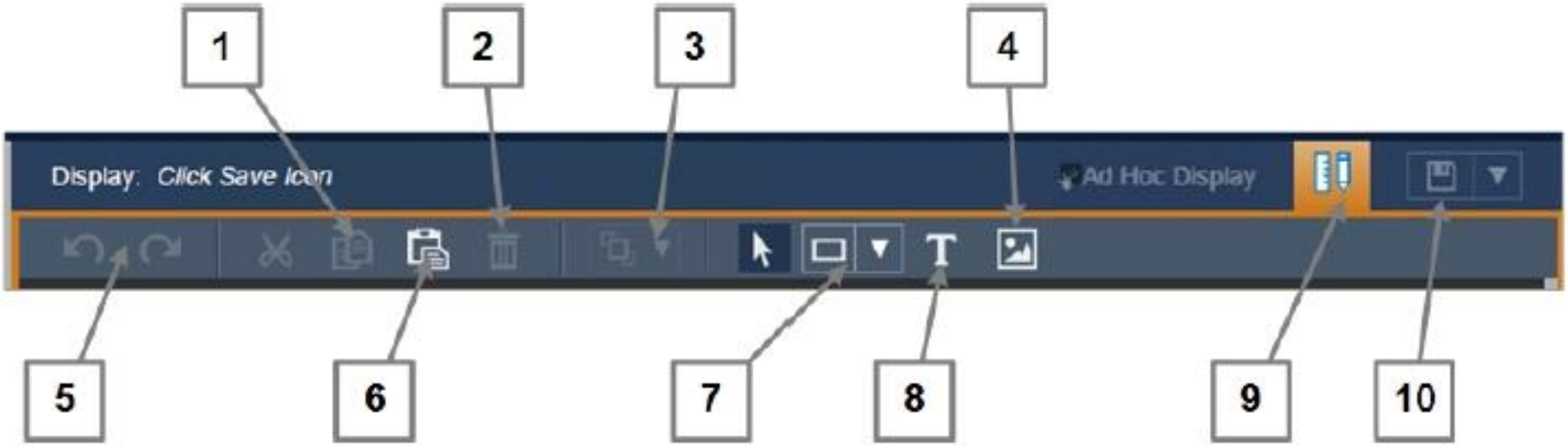
PI Vision Symbols

Trend	Trends can show the value of one or more data items over a time period. Trends are typically used to display time series data, though they may also include non-time series data. Upon exiting Design mode, you can lick to view trend cursors, pan across the time range, zoom in and out and hide traces. Right click to configure the value scale or remove traces.	Multiple
Value	A value is the reading or snapshot obtained for a data item at the end time of the display. It is shown as a number, time stamp, string, or digital state. Right click to format how the value is displayed or to add Multi-State.	Single
Table	The table symbol contains columns that include the name, value, description, and other summary data about a data item. These summary data values take their intervals from the display's time range as defined in the time bar. Right click to configure table columns.	Multiple
Vertical Gauge Horizontal Gauge Radial Gauge	These three symbols are identical in every way, except their orientation. The zero and span of the symbol are from the PI point attributes. If the data item is a AF attribute of formula type, the minimum and maximum traits on the attribute are used. Right click to format the gauge or to add Multi-state.	Single
XY Plot	An XY Plot shows a correlation between one or more paired sets of data. On an XY Plot (also called a scatter plot), the X scale shows possible values for one of the items in the pair and the Y scale shows the value of the other item in the pair.	Multiple
Asset Comparison Table	The asset comparison table symbol allows you to compare measurements from similar types of equipment by organizing your data by assets. Each asset is assigned its own row while columns contain the asset's selected attributes.	Multiple

Tools and Symbols

Static Shapes	Static shapes add rectangles, circles, lines, arcs or polygons to the display. Right click to format the shape or add Multi-state.
Text	Add a line of text to the display. Add a hyperlink to the text and search for an existing display to link to. Right click to format the text or add Multi-state.
Image	Add an image to the display. Supports most file formats including JPG, TIF, GIF (Static and animated), BMP, and SVG. The maximum image size is 2 MB.
Arrange	To arrange multiple objects by aligning them or bringing one of them backward or forward, click the Arrange button on the editing toolbar. There are many options for arranging or aligning display objects, including sending an object forward or back, aligning multiple objects and distributing objects on the display.

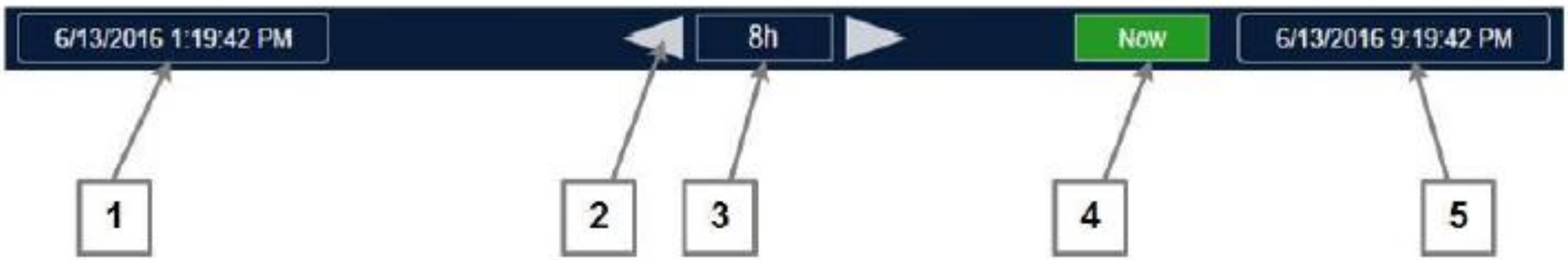
Display



Display


1
2
3
4
5
6
7
8
9
10
PI Vision
Assets
Search in Production Line1
Home
OSIsoft Plant
Production Area
Production Line1
Mixing Tank1
Storage Tank1
Attributes
Mixing Tank1
Future Data
Level_Forecast
Process Variables
External Temperature
Flow Rate
Internal Temperature
Level

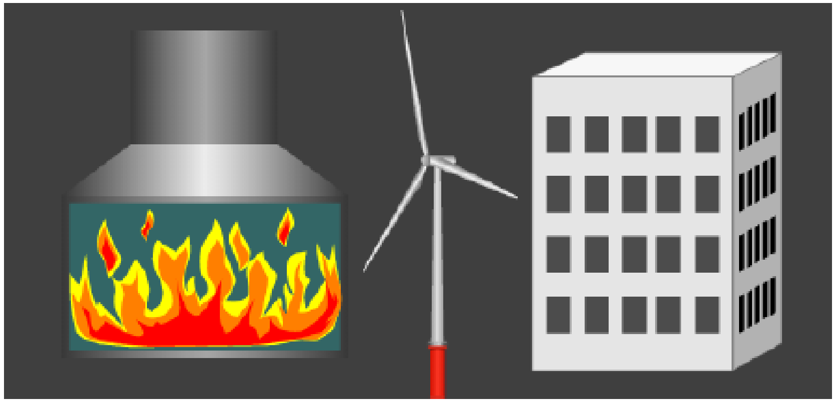
Time

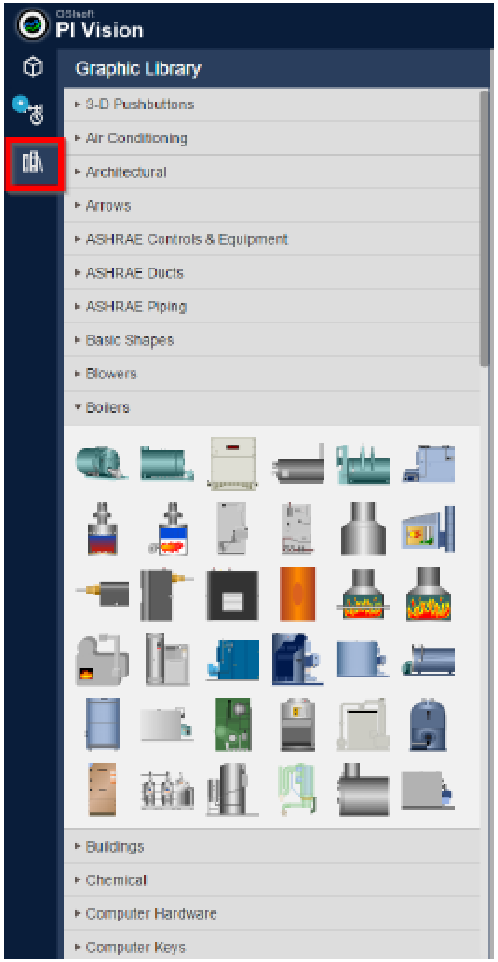


Graphics Library

A large selection of graphics is available in the Graphic Library pane. The graphics are organized inside stencils belonging to a wide range of categories and industries. Many of the graphics have characteristics such as color, fill type, orientation, or background, which you can modify.

To open the Graphic Library pane, click on the Graphic Library tab, located below the Events tab.

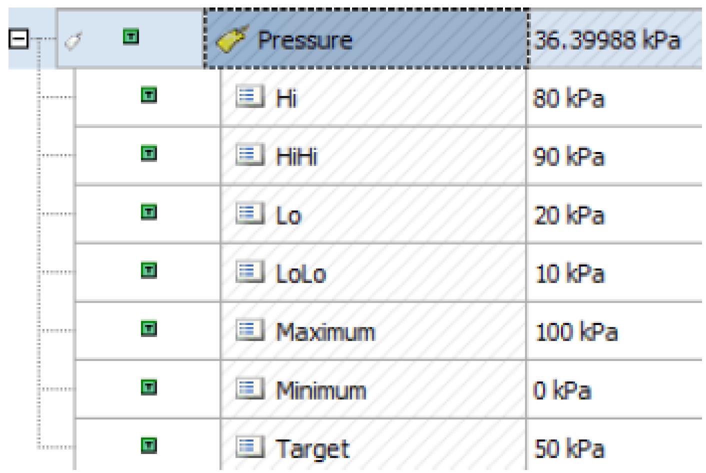


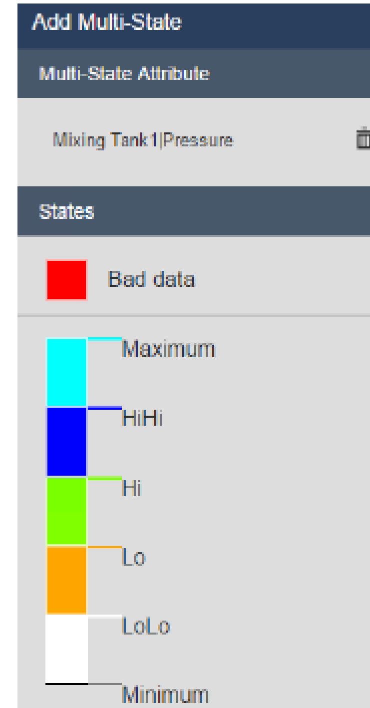


Multi-State Limits

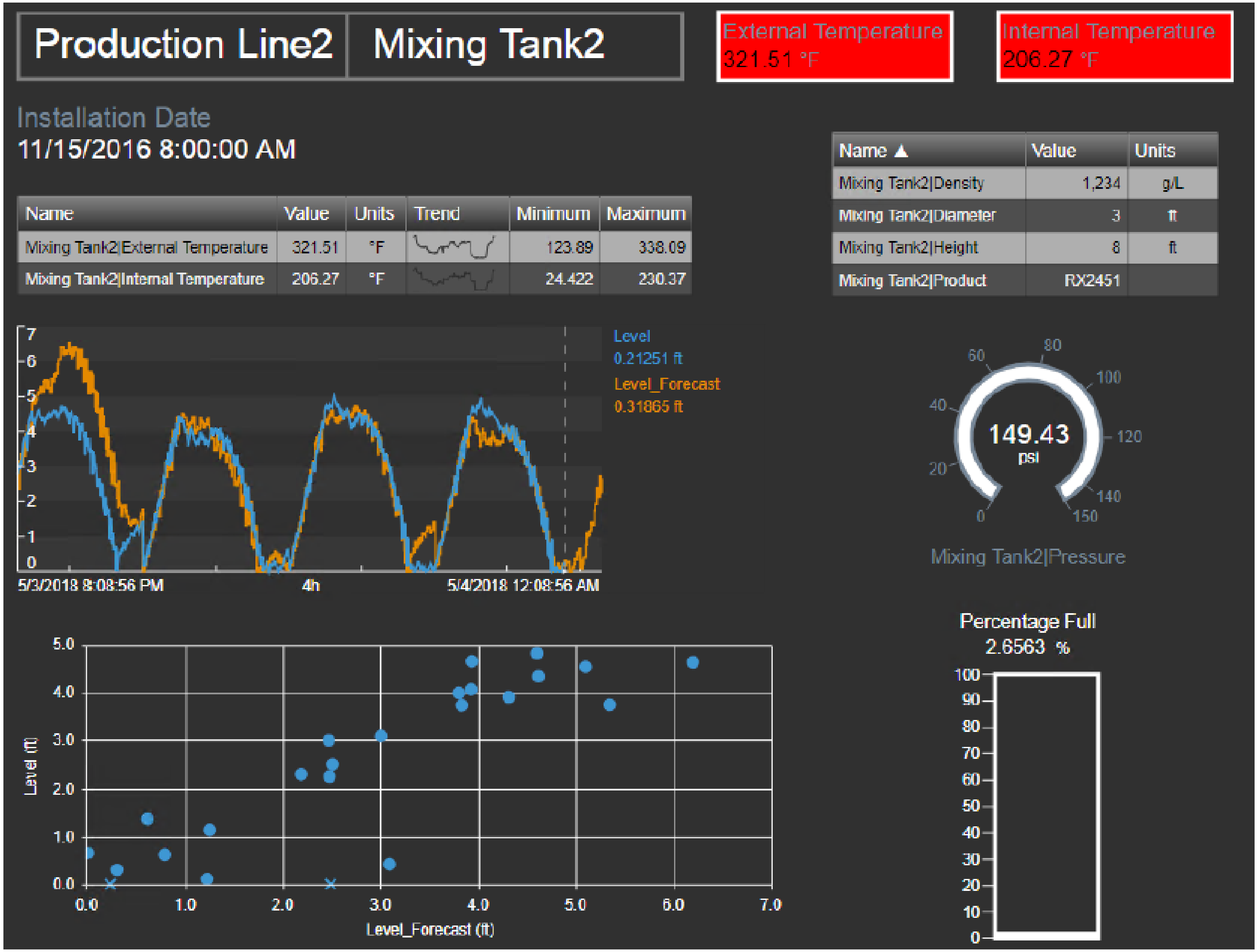
Notice that you cannot change the Multi-State limits for Mixing Tank 1|Pressure

If the AF attribute has assigned Limits (which were introduced in AF 2016) then the multi-state will use the limits defined in AF and the user will not be able to change them, they will only be able to change the colors associated with each state. Pressure has been configured with AF Limits which are child attributes with the corresponding limits property:



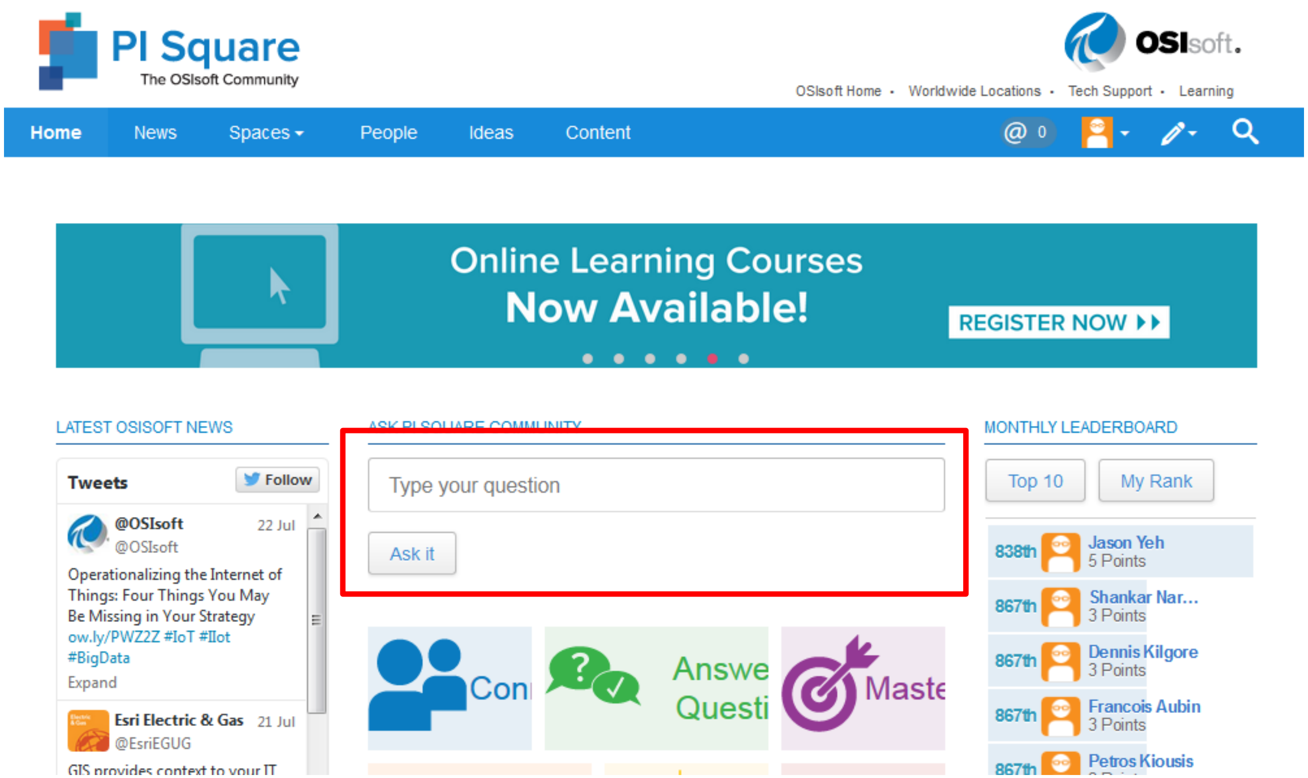


Example



Production Line2
Mixing Tank2
External Temperature 321.51 °F
Internal Temperature 206.27 °F
Installation Date
11/15/2016 8:00:00 AM
Name | Value | Units | Trend | Minimum | Maximum
Mixing Tank2|External Temperature | 321.51 | °F | 123.89 | 338.09
Mixing Tank2|Internal Temperature | 206.27 | °F | 24.422 | 230.37
Name | Value | Units
Mixing Tank2|Density | 1,234 | g/L
Mixing Tank2|Diameter | 3 | ft
Mixing Tank2|Height | 8 | ft
Mixing Tank2|Product | RX2451
Level
0.21251 ft
Level_Forecast
0.31865 ft
5/3/2018 8:08:56 PM
4h
5/4/2018 12:08:56 AM
149.43
psi
Mixing Tank2|Pressure
Percentage Full
2.6563 %
Level (ft)
Level_Forecast (ft)

PI Square



Make friends!

Connect and consult with the OSIsoft community in PI Square, where you can share knowledge with other users of the PI System (OSIsoft employees, partners, customers).

감사합니다

谢谢

Danke

Gracias

Merci

Thank You

ありがとう

Спасибо

Obrigado